# Supplementary Material: More Than the Risk: Frontier LLM Behavior in AI Development Races Depends on the Rival

Anonymous Author(s)
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## ABSTRACT

This supplement supports More Than the Risk: Frontier LLM Behavior in AI Development Races Depends on the Rival. It records the quality-control screen, the theory comparison, the scripted-rival evidence, and the human diversity analysis, together with prompt and game robustness checks. These materials use the same routes, task and evidence boundaries as the main paper and do not extend its claims beyond the idealised race.

## KEYWORDS

Large language models, AI development races, multi-agent systems, repeated games, AI safety, opponent-conditioned behaviour

## HOW TO READ THIS DOCUMENT

Each section stands on its own and is named for what it measures. Section letters restart at A because this is supplementary matter.

- **The quality-control screen.** What the screen asked, when its rule was fixed, which of the nine routes cleared it and on which condition, and the three facts that bound what a verdict may be used for. The screen is recorded here as an entry condition; its full measurement interpretation is outside the behavioural contribution.
- **Human-benchmark replication check.** Whether the reference human study's reported pattern reproduces on the released participant data under our own analysis code.
- **Claim-scoped citation audit.** For each cited work, the claim it is used to support and the claim it does not support.
- **Risk preference and assigned personas.** The persona families, and how a prompted persona compares with measured human risk preference.
- **Selection-strength sweep of the evolutionary benchmark.** Whether the gap between theory and the audited routes depends on one parameter setting.
- **Crossed representation-robustness design.** The narrative skin by symbol mapping rerun, completed on both routes selected for the crossed representation study, and what two routes do and do not license.
- **Predictive structure of unsafe choices.** The feature analysis behind the drivers reported in the main paper.
- **Two-player descriptive baseline, full scope.** Every checkpoint and condition, including those the main paper does not plot.
- **$N$-player self-play scope.** Group sizes three to five.
- **Trajectory embedding, by population and by outcome.** The same embedding twice, coloured by population and then by UNSAFE rate.
- **Cluster-count sensitivity and trajectory diversity.** How the human archetypes and the diversity comparison behave as the analysis choices move.
- **UNSAFE rate by persona band and relative position.** Cells reported only here, with observed rather than assigned position.
- **Joint outcome dynamics.** The race read as a walk on the three joint states, one transition matrix per route, and the two roads back to mutual restraint.
- **The designed boundary at round five.** Whether play moves where the stated continuation probability drops, and why that contrast is across round positions rather than a discontinuity estimate.

Four terms are used with fixed meanings throughout. A *route* is the exact provider string a request is addressed to, for example `google/gemini-3-flash-preview`. An *endpoint* is a route that carries an admission verdict from the probe battery below. A *checkpoint* is one dated model version, which is what a comparison across the rows of a behavioural table compares. *Model* is used only generically, for a language model as a kind of system. Where a number belongs to a route rather than to a model in general, the route string is printed beside it.

## A EXTENDED RESULTS AND ROBUSTNESS CHECKS

This supplement collects the material referenced from the main paper as supporting rather than extending the main claims: the record of the quality-control screen and which routes it admitted (§A.1), the human-benchmark replication check (§A.8), the persona comparison (§A.10), the predictive analysis (§A.13), and the full two-player and $N$-player descriptive results (§A.14, §A.15). It also carries the evidence the main paper only summarises: the separate and larger battery on the one model with published weights (§A.2), the selection-strength sweep of the evolutionary benchmark (§A.6), the crossed representation-robustness design completed on both routes selected for the crossed representation study (§A.7), the matched group-size comparison (§A.23), the scripted-opponent campaign (§A.26), the cluster-count sweep behind the human archetypes (§A.20), the trajectory-diversity statistics (§A.21), the pilot roster the projections are built on (§A.11), the joint outcome dynamics of a race (§A.24), and the designed horizon boundary at round five (§A.25).

### A.1 The quality-control screen: what it asked, and which routes passed

This subsection records the screen as an entry condition and nothing more: what it asked, when its rule was fixed, which routes cleared it, and the three facts about it that bound what a verdict may be used for. It is deliberately short. Nothing in this supplement ranks routes by their probe scores or treats the verdict as a measurement of capability.

The screen is a frozen bank of 20 probes over six domains, asked three times each, giving 60 scored answers per route and 540 in all. The six domains are recalling the rules, reading the stage-payoff matrix, rebuilding the current state from a short history, taking one state transition, scoring a finished race, and computing a closed-form expected payoff. A route is admitted only if it clears all three of the conditions recorded in the caption of Table 1: 80% of all its answers, 75% within state reconstruction and 75% within terminal scoring. Expected payoff is recorded and never blocks admission, because the game asks an agent to choose an action rather than to solve for an expected value.

Two properties of the rule are what let it serve as an entry condition. The bank, the three conditions and their values were fixed and recorded before the campaign ran, in the collection protocol under `docs/`. And the screen scores probe answers against ground truth without ever reading a race, so no gameplay result can reach a verdict, whatever order things were run in. We do not call the screen pre-registered, because three of the nine routes received their first verdict after their own gameplay had completed; what is checkable, and what the defence rests on, is the fixed bank, the fixed rule and the fact that the instrument cannot see the outcome it selects on.

Three facts make the nine rows of Table 1 comparable with one another. All nine ran under one protocol identifier. All nine answered the same probe bank, whose SHA-256 hash begins `2953fb47`, and read the same rules context, whose hash begins `b80981a5`; the analysis script fails closed and refuses to tabulate a route whose hashes differ. Eight routes ran at benchmark task version 7 and one, `google/gemini-3.5-flash-lite`, at version 8, whose outgoing request omits the reasoning-budget argument because that provider returns an invalid-argument error when the argument is present at all. That is a difference in the transport contract rather than evidence about the model, it is recorded in that route's own manifest, and the amendment introducing per-route contracts is dated in the collection protocol. For every other route the outgoing request is byte-identical to the earlier runs. One failed run is retained rather than deleted: the same route at task version 7 ended in exhausted transport retries and produced no scoreable answer, so it appears nowhere in the table and is kept as a transport failure beside the campaign.

*The verdict does not turn on where the thresholds sit.* Each declared value carries 5.0 points of headroom, and the three ranges hold together rather than one at a time: the admitted set is the same at all 676 threshold combinations inside them. In the other direction two of the three conditions can be loosened to zero without admitting anyone new, because state reconstruction refuses the other four routes on its own. That domain is the one exception, and it is worth stating rather than averaging away: one achievable step down on state reconstruction, from 75% to the 73.3% the fifteen-answer grid allows below it, admits `google/gemini-3.1-flash-lite-preview` and nobody else.

*The battery has been administered three times.* All three administrations used the same bank and the same rules context at temperature zero. Between the two with no recorded contract difference, two of the four routes present in both moved, each by two answers in fifteen on state reconstruction, and the other two did not move at all. No verdict changed. Across the earlier change from a 128-token output cap to 256 one verdict did move, and it is reported because that route is admitted: Claude Sonnet 5 was refused under the 128-token cap and admitted under 256. A truncated answer is scored wrong and a low cap can truncate, so that comparison is not a clean repeat and we do not present it as one.

*Two admitted routes pass by nothing at all.* State reconstruction and terminal scoring are each scored over fifteen answers, so a 75% threshold asks for 11.25 correct and the lowest score that can clear it is 12 of 15, or 80.0%. GPT-5.5 clears state reconstruction at exactly that score and Claude Sonnet 5 clears terminal scoring at exactly that score, so both are admitted with a margin of zero. One answer otherwise would put GPT-5.5 at 73.3%, which is the score on which Gemini 3.1 Flash-Lite is refused. An admitted route's pass margin can therefore be narrower than the movement the same instrument shows between two identical administrations. That is why a verdict is used here as a declared entry condition and never as a trait estimate, and why no analysis in this paper interpolates between probe scores, ranks admitted routes by probe accuracy, or reads the admitted against refused split as a measured contrast.

*The split coincides exactly with the tier word in the route name.* Calling a route small when `mini`, `nano` or `lite` appears as a whole word of its name splits these nine endpoints into exactly the two sets the screen produced: all four refused routes carry such a word and none of the five admitted routes does, nine of nine. Enumerating all $C(9,4) = 126$ ways of relabelling which four names carry a size word gives an exact one-sided permutation $p$ of $1/126$, or 0.008. This does not show that the battery measures nothing, because comprehension that simply tracked model scale across these nine endpoints would look the same and nine routes cannot tell the two apart. What follows is narrower and is the reason the screen is used only as an entry condition here: on this roster the screen draws no line that the route name does not already draw, so no claim resting on the admitted against refused split is separable from a claim about scale. Breaking that tie needs endpoints whose name and whose comprehension disagree, and this roster contains none. A broader roster is required before the screen can support a claim that is separable from endpoint scale.

*Nothing was discarded.* All nine routes played the same confirmatory baseline with the same race counts, and every refused route's behaviour is reported in this supplement beside the five admitted ones. Four further advertised routes never produced a scoreable answer at all, because transport failed before any probe was sampled; they were retained as failure records and never replaced by a substitute endpoint.

## A.2 A second, larger battery on the one model with published weights

One route in this study is not a hosted endpoint and is not one of the nine. Its weights are published, so decoding is under our control rather than a provider's, and it is the route that carries the paired decision-card diagnostic of the main paper. It was screened by its own instrument, a battery of 41 probes over the same six domains, and that battery is reported here so that the two are never mistaken for one another. Nothing in this subsection enters

**Table 1: The nine screened routes on the three conditions that decide admission, and which condition refused each route that did not enter. A route is admitted only when all three clear: overall accuracy at least 80%, state reconstruction at least 75% and terminal scoring at least 75%. Every cell is the percentage of that domain's probes answered correctly. Overall accuracy rests on all 60 scored answers per route, state reconstruction and terminal scoring on 15 each, so those two move in steps of one fifteenth, about 6.7 points. Expected payoff is recorded as a diagnostic and gates nothing; no route reaches 75% on it. The per-domain results in full are retained in this supplement.**

| Exact route | Overall accuracy | State recon. | Terminal scoring | Verdict |
|---|---|---|---|---|
| `google/gemini-3-flash-preview` | 93.3 | 93.3 | 100.0 | admitted |
| `anthropic/claude-opus-5@default` | 91.7 | 93.3 | 100.0 | admitted |
| `openai/gpt-5.4-2026-03-05` | 90.0 | 100.0 | 100.0 | admitted |
| `openai/gpt-5.5-2026-04-23` | 90.0 | 80.0 | 100.0 | admitted |
| `anthropic/claude-sonnet-5@default` | 85.0 | 100.0 | 80.0 | admitted |
| `google/gemini-3.1-flash-lite-preview` | 80.0 | 73.3 | 80.0 | refused, state recon. |
| `openai/gpt-5.4-mini-2026-03-17` | 75.0 | 66.7 | 86.7 | refused, two conditions |
| `google/gemini-3.5-flash-lite` | 65.0 | 60.0 | 60.0 | refused, all three |
| `openai/gpt-5.4-nano-2026-03-17` | 51.7 | 20.0 | 66.7 | refused, all three |

a strategic claim, and no score here is pooled with a score from the nine-route screen: different probe bank, different route, different number of repetitions.

The model is Qwen2.5 7B Instruct at half precision. Probes were put three ways rather than one: unaided, reworded into a paraphrase that keeps the question and changes the sentence, and accompanied by a disclosed calculation of the quantities the question needs. Each probe was answered five times in each of the three, at temperature zero under one fixed base seed, and the seven probes that list their answer options were additionally asked with those options in the reverse order. That gives 685 retained probe answers, every one of which the semantic scorer could read.

What the battery shows is a gap between stating a rule and using it. Table 2 reports it by domain. Recalling the rules and reading the stage-payoff matrix are near perfect unaided, and everything that requires carrying the state forward or doing the arithmetic collapses: unaided, this route rebuilds the current state on about a quarter of its answers, and takes a single state transition or computes a closed-form expected payoff on none of them. Disclosing the arithmetic lifts the whole battery from 52.1% to 75.6%, which measures the use of a disclosed result and not unaided comprehension. Two further readings follow. Unaided, every one of these scores sits below the matching condition the nine-route screen declares, and while the two instruments differ enough that this is a description rather than a verdict, it is the reason this route supplies a presentation diagnostic and nothing else. With the arithmetic disclosed the route reaches the terminal-scoring condition exactly and still clears neither of the other two. And strict format compliance, 32.1% over the whole battery, is far below the semantic parse rate, which was 100%, so on this route the format a reader sees and the answer the model gives come apart almost two times in three.

The claim this battery can carry is narrow and worth stating as the analysis records it: probe accuracy supports rule and arithmetic performance under the tested prompt pool, it does not establish an internal world model, and accuracy with the calculation disclosed measures the use of disclosed outputs rather than unaided comprehension. The gameplay beside it is the paired diagnostic itself, 30 races and 558 decisions in each of the two presentations with no parse failure, and it is read only as a within-route contrast between two presentations of one mechanism.

**Table 2: The 41-probe battery on the one model with published weights, Qwen2.5 7B Instruct, by domain. "Unaided" pools the plain and the reworded askings and both option orders; "disclosed" is the same probes accompanied by the arithmetic they need. Every cell is the percentage of answers the semantic scorer marked correct, over the number of answers shown. This battery is a separate instrument from the nine-route screen and its scores are never compared with those in Table 1.**

| Domain | Unaided | $n$ | Disclosed | $n$ |
|---|---|---|---|---|
| Rule recall | 96.4 | 140 | 100.0 | 50 |
| One-stage payoff lookup | 100.0 | 40 | 100.0 | 20 |
| State reconstruction | 27.8 | 90 | 55.6 | 45 |
| State transition | 0.0 | 60 | 66.7 | 30 |
| Terminal scoring | 45.5 | 110 | 75.0 | 40 |
| Expected payoff | 0.0 | 40 | 50.0 | 20 |
| All probes | 52.1 | 480 | 75.6 | 205 |

## A.3 Earlier two-route confirmatory baseline

The new neutral gameplay baseline used protocol v3, the same mechanism and prompt for the two routes used in the direct EGT comparison, ten repetitions per risk level, and 30 races per route. Each route produced 558 structured decisions with zero parse failures. The entries in Table 3 are decision-weighted UNSAFE rates; they are not pooled across routes and do not estimate an evolutionary equilibrium.

**Table 3: Two-route confirmatory block used for the direct EGT comparison. Each route has 30 races, 558 decisions, and zero parse failures.**

| Exact route | $r = .1$ | $r = .6$ | $r = .9$ | Races |
|---|---|---|---|---|
| `google/gemini-3-flash-preview` | 98.9% | 73.1% | 60.2% | 30 |
| `anthropic/claude-sonnet-5@default` | 89.2% | 48.4% | 32.3% | 30 |

The Gemini manifest records that the hosted SDK stripped the requested sampling seed. The Claude manifest records that the seed was forwarded but provider-side application was not confirmed. Temperature was requested as 0.7 for both routes but was not forwarded by this SDK path. These facts bound the result to the exact hosted route and software contract.

To make the comparison boundary explicit, Figure 1 separates the risk response, the route-specific departure from the strong-selection reconstruction, the nearest-rule diagnostic, and its mismatch rate. This is an analysis artifact rather than an equilibrium test: the frontier routes generate prompted self-play trajectories, whereas EGT samples stationary population composition.

Which strategy takes that population is a one-line result, now shown in Figure 2. Read off the archived EGTtools fixation probabilities in `egt_frontier_invasion.csv`, the strategy that dominates the independently simulated finite-mutation chain ($\beta = 2$, $\mu = 0.02$) switches with risk: Always UNSAFE at $p_r^{\max} = 0.1$ with 91.9% of the chain, the conditional rule that opens UNSAFE at 0.6 with 96.3%, and the conditional rule that opens SAFE at 0.9 with 93.9%. Exactly nine of the twelve off-diagonal invasions per risk level clear neutral drift $1/Z = 0.01$, and the smallest fixation probability among those nine is 0.997, so the ordering is not marginal. Two boundaries travel with that sentence. The fixation probabilities are evaluated in the EGTtools small-mutation limit while the shares come from the finite-mutation chains, which are two different simulation layers; and no language-model route plays this population process, so none of it is a latent-strategy statement about any endpoint.

## A.4 The shape of a policy behind a rate

Every gameplay number in §A.3 is a rate: the share of decisions that came out UNSAFE. A rate is silent about the thing an iterated game is for. Two routes can sit at the same rate while one is reading the board and the other is reciting a constant, and no table in this supplement separates those two cases. Figure 3 is the object that does, and that is what it adds to the neutral baseline: not another number for how much UNSAFE play there was, but the policy that produced it.

The read-out is the memory-one fingerprint of the evolutionary game theory literature: the probability of playing UNSAFE given the pair of moves that preceded it, the route's own move first and its rival's second. The choice is not an analytical convenience. The prompt shows the players only the immediately preceding round, so memory-one is the whole state the game makes available, and four conditional probabilities are therefore a complete description of a stationary policy rather than a summary of one. The fingerprint is computed over the same nine-route neutral baseline the rest of this paper reports, 270 races and 5,022 decisions, of which the 4,482 that have a predecessor round enter the calculation; round one is excluded rather than folded in, because an opening move is a different object from a reply.

The two lagged effects beside it are Mantel-Haenszel differences, stratified on the risk level and on the other lagged move. Both stratifications are load-bearing. Risk has to be held fixed because the three risk cells have very different base rates, so pooling them would let a route that simply plays more UNSAFE at low risk look responsive. The other lagged move has to be held fixed because these are self-play races: both companies are the same route, so an unstratified reading of what a route does after its rival went UNSAFE would largely measure the route's own phase. Intervals resample races within their risk cell, the same clustering every other interval in this paper uses.

Reading the two panels together separates three kinds of route that a rate column would print side by side. Six of the nine answer their rival's last move by moving toward it, and every route whose terms are defined at all has a negative own term, so the pattern is not momentum in either direction but matching the rival while correcting itself. On GPT-5.4 mini that own term is the one whose interval spans zero, −4.8 pp [−12.7, +3.4], so seven of the eight are estimated away from zero and the eighth is not. Two routes, GPT-5.4 mini and GPT-5.4 nano, carry a rival term pointing the other way; they are also the two whose UNSAFE rate barely moves with the stated risk. Claude Opus 5 is the third kind: a step whose two seats never diverged in 279 rounds, for which the question the fingerprint asks does not arise.

*What it does not show.* Responding to a rival is not understanding a rival, and nothing here licenses a claim about strategy, intent or reasoning: these are conditional frequencies in one game at one prompt version, on nine commercial endpoints that are not a sample from a population of models. The own term in particular is not a preference. After its own UNSAFE move a route is also carrying more accumulated private risk, which is a real feature of the game rather than a confound, but it means the term measures a response to the state and not to the move alone.

## A.5 How much of a route's number is the run?

Every route-level rate in this paper comes from one thirty-race run, so a reader is entitled to ask how much of a rate is the route and how much is the particular run. One route answers that question directly. Sampling on these endpoints is not reproducible: the SDK strips the seed request before it reaches the provider, which every run manifest records as `not_applied_sdk_stripped_for_route`. Two runs of the same cell are therefore genuinely independent draws from the route, not a rerun of one draw.

`google/gemini-3-flash-preview` was collected twice under the frozen protocol `ai-race-frontier-baseline-v3`: at task version 3 on 2026-09-08, which is the run reported everywhere else in this paper, and again at task version 4 on 2026-09-09 on a different billing identity. The repeat was not planned as evidence. It rode along with the push that carried the reasoning-budget amendment of §A.1 and was recovered from the Kaggle archive afterwards. Protocol, prompt hash, mechanism, seed structure and the full set of thirty (game seed, risk) blocks are identical across the two runs, and `scripts/analyze_baseline_replication.py` refuses the comparison unless they are.

Agreement is close. UNSAFE rates move from 98.9% to 100.0% at $p_r^{\max} = 0.1$, from 73.1% to 74.2% at 0.6, and from 60.2% to 59.7% at 0.9: the largest per-risk difference over 186 decisions is 1.1 percentage points, and the risk response, the quantity the admission analysis in §A.1 actually uses, moves from 38.7 to 40.3 percentage points. That is small beside the between-route spread of 7.0 to 100.0 points that the same analysis reports, which is the comparison this check exists to protect.

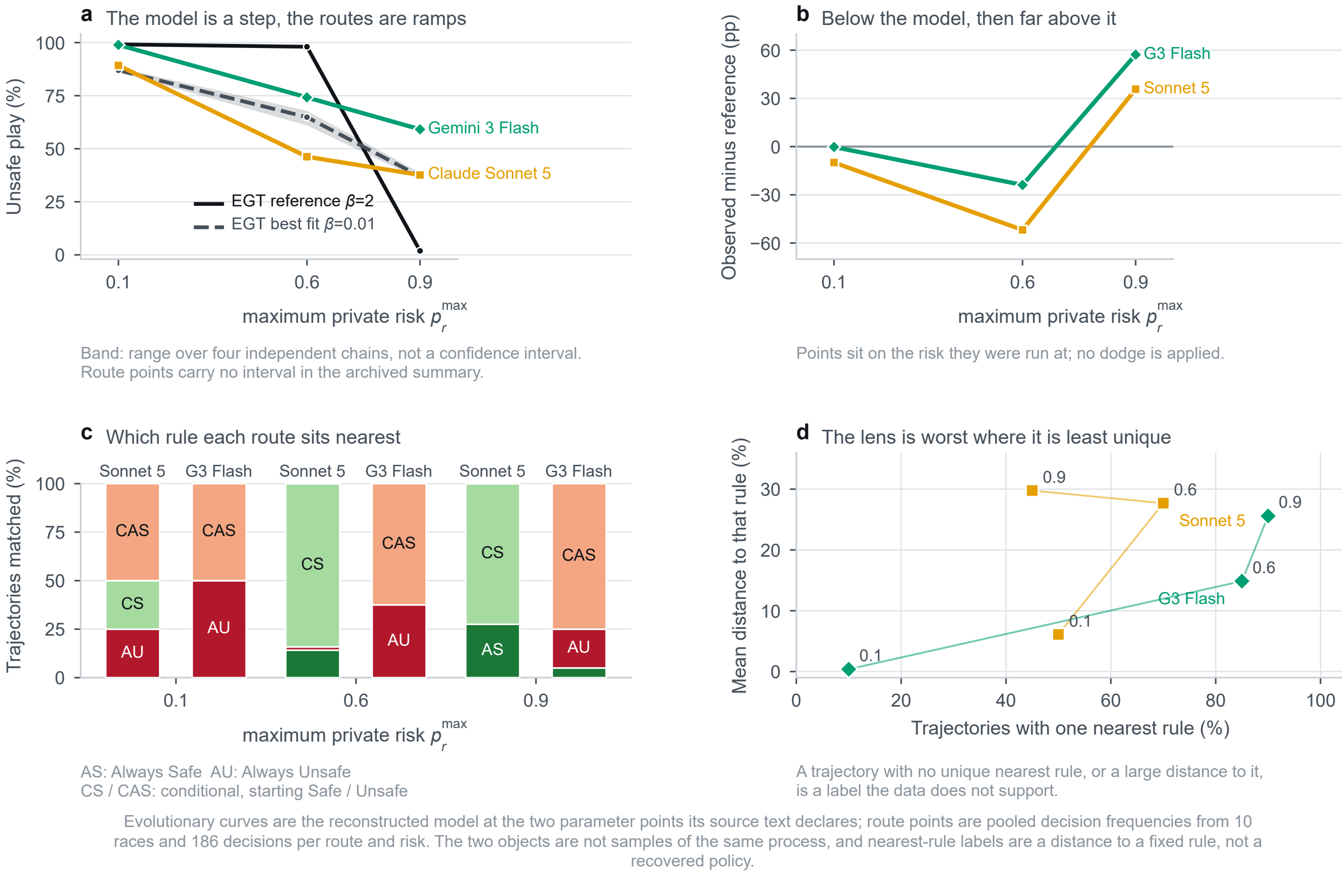



**Figure 1: The reduced evolutionary model against the two routes that were run through the comparison protocol, `anthropic/claude-sonnet-5@default` and `google/gemini-3-flash-preview`. These are two of the five routes the comprehension gate admits, not the admitted set. (a) At its reference parameter point the model is a step in risk, holding UNSAFE play near the ceiling through $p_r^{\max} = 0.6$ and dropping it to 1.9% at 0.9; both routes are ramps and both stay far above the model at 0.9. The band on each model curve is the range across four independent seeded chains, which is a convergence diagnostic and not a confidence interval; the route points carry no interval because the archived summary holds none. (b) The same contrast signed against the reference, with each point on the risk it was run at. (c) Which of the four rules each route's trajectories sit nearest, one stack per route. (d) Why (c) is a lens and never a finding: the labels are least unique exactly where the distance to the nearest rule is largest.**

It is one route and one repeat, so it bounds run-to-run variation for this route at this sample size and licenses nothing for the routes that were collected once. The repeat is stored at `results/frontier/baseline_replication/`, deliberately outside the campaign tree: both campaign analysers key their results by model route, so a second run of an already-represented route placed inside that tree would not raise an error, it would quietly displace the reported one.

## A.6 Selection-strength sweep of the evolutionary benchmark

The comparison in §A.3 uses one setting of the reconstructed evolutionary model. A reviewer may reasonably ask whether the gap between that model and the audited routes is a property of the model or an artifact of that one setting. This subsection answers that question by sweeping the setting.

Two numbers govern the model. *Selection strength* $\beta$ says how strongly a simulated agent prefers to copy a better-performing neighbour: at $\beta$ near zero copying is almost random, and at large $\beta$ it is almost deterministic. The *mutation rate* $\mu$ is the chance, per update, that the sampled individual abandons its strategy and switches uniformly to one of the other three. The sweep covers ten values of selection strength, $\beta \in \{0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 1, 2, 3, 10\}$, crossed with three rules for setting the mutation rate: $\mu = \beta/Z$, the main-text convention, which gives $\mu = 0.02$ at $\beta = 2$; $\mu = 1/Z = 0.01$; and a fixed $\mu = 0.05$, the convention under which the source study reports its best fit. With $Z = 100$ individuals, four strategies, and the mechanism's three maximum-private-risk levels (0.1, 0.6, 0.9), the sweep contains 90 cells built from 29 distinct parameter

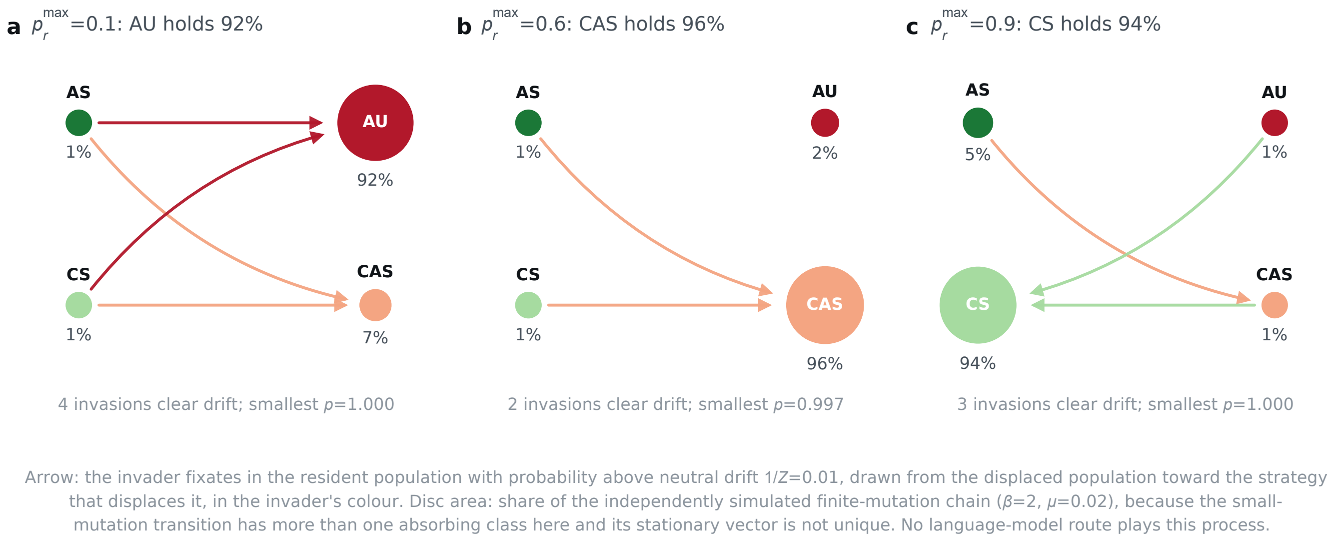



**Figure 2: Risk-conditioned invasion topology in the reconstructed evolutionary game. Each panel is a standard invasion graph at one maximum private-risk level. The four nodes are Always Safe (AS), Always Unsafe (AU), conditional Safe (CS) and conditional Always Unsafe (CAS); node area shows the share of the independently simulated finite-mutation population. An arrow points from the resident strategy to an invader whose EGTtools fixation probability exceeds neutral drift $1/Z = 0.01$. The graph shows which strategy can replace which resident, while node areas show the finite-mutation population composition.**

pairs, since some pairs coincide across rules and were simulated once. Each cell is estimated from four independent chains of 400,000 steps with a burn-in of 100,000, thinned by 100, giving 4,000 retained samples per chain.

The reference setting reproduces. At $\beta = 2$ with $\mu = \beta/Z = 0.02$, the sweep returns a decision-weighted UNSAFE share of 99.2%, 98.0%, and 1.9% at maximum risks 0.1, 0.6, and 0.9: the same three-point profile the main paper attributes to the published reference, matched to within 0.0002, 0.0001, and 0.0001 in share, far inside the 0.02 tolerance the check declares in advance. The sweep uses a deliberately disjoint seed scheme from the reproduction script, so this agreement is an independent-seed check rather than a rerun of the same random numbers.

The qualitative gap does not close anywhere on the grid at the reference selection strength, but it does close at very weak selection. Of the 90 cells, four reproduce the qualitative shape the admitted endpoints show, namely near-total UNSAFE play at risk 0.1, a middling rate at 0.6, and a lower but clearly non-zero rate at 0.9, declining at every step. Three of those four also pass the mixing diagnostic. All of them sit at weak selection: $\beta = 0.01$ and $\beta = 0.03$ under $\mu = 0.05$, and $\beta = 0.003$ and $\beta = 0.01$ under $\mu = 1/Z$. Ten of the 90 cells are flagged as mixing-suspect, meaning their four independent chains disagreed by more than 0.05 on the aggregate UNSAFE share, so their chain mean must not be read as a stationary distribution; almost all of them are the very small mutation rates that $\mu = \beta/Z$ produces at small $\beta$, which make the four uniform-strategy states nearly absorbing.

The best-fitting cell is the same for both audited routes, and it is not the reference setting. Fitting is by root-mean-squared deviation, in percentage points, between the model's three-point UNSAFE profile and the route's observed decision-weighted UNSAFE rates at the same three risk levels; the largest single-condition deviation is reported beside it so that one badly missed condition stays visible. For both routes the minimum-error cell is $\beta = 0.01$ with fixed $\mu = 0.05$, whose profile is 87.3%, 63.9%, and 38.0%. Against `anthropic/claude-sonnet-5@default`, observed at 89.2%, 48.4%, and 32.3%, that cell has an error of 10.28 percentage points and a largest single-condition miss of 17.70 points, at risk 0.6. Against `google/gemini-3-flash-preview`, observed at 98.9%, 73.1%, and 60.2%, the error is 15.13 points and the largest miss 21.14 points, at risk 0.9. Even the best cell on the grid is therefore more than ten percentage points away from either route.

*All-five admitted-route extension.* The preceding block is the earlier two-route confirmatory comparison. To check whether its qualitative reading depends on that restricted roster, we fit the same existing 90-cell sensitivity table to the canonical neutral-baseline profiles of all five admitted routes. This is analysis only: no LLM requests or evolutionary chains were added. Distances are RMSEs in percentage points over the three matched risk levels, and the primary cells below are required to be well mixed. The best weak-selection cell is the same in every route, $\beta = 0.01$ with fixed $\mu = 0.05$; the four graded routes are substantially closer to it than to the reference $\beta = 2$ (Table 4), while Claude Opus 5 remains a qualitative near-step exception.

One boundary is worth restating, because the reproduction lane states it itself. This is a repo-native reconstruction of the disclosed reduced evolutionary model, not a bitwise reproduction of the original authors' code: that code, the generated payoff matrices, the payoff Monte Carlo seeds, and the exact library revision are not public in the preprint version we worked from, and the compiled comparison class could not be executed on this interpreter because

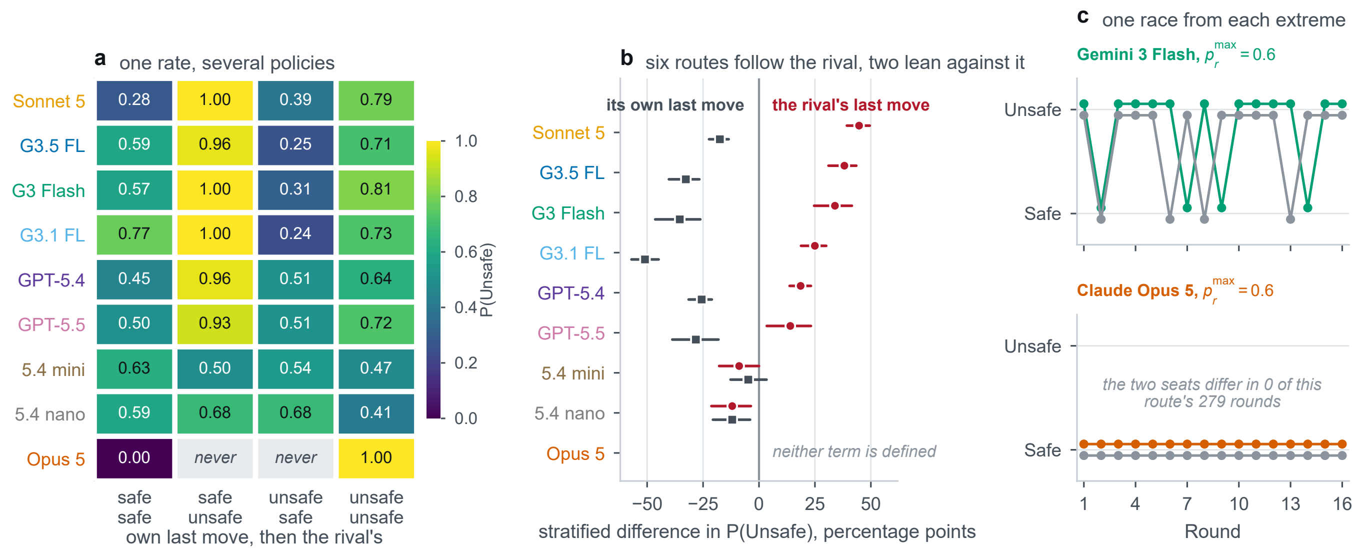



**Figure 3: The memory-one policy behind each route's UNSAFE rate, over the nine-route neutral baseline. (a) One row per route and one column per preceding pair of moves, the route's own first and its rival's second, with the conditional probability printed in every tile; row labels carry the route colours used throughout the paper, so the panel needs no legend. Two of Claude Opus 5's tiles are marked absent rather than left blank: its two seats never once played differently, so the mixed contexts did not arise and there is nothing there to measure. (b) The two lagged associations on one scale, Mantel-Haenszel differences in $P$(UNSAFE) stratified on the risk level and on the other lagged move: how far UNSAFE play differs after the rival's UNSAFE move, and how far it differs after the route's own, with race-clustered 95% intervals and the routes in the same order as panel a. (c) Two races at the same risk level under the same protocol, one from a route that answers its rival and one from the route that does not move at all, with the two seats drawn separately. Every quantity here is an association within a stratum and not a causal effect. Nothing is randomised: these are conditional frequencies from nine commercial endpoints in one game at one prompt version, self-play throughout, and the own term in particular also carries the accumulated private risk that an earlier UNSAFE move leaves behind, so it is a response to the state and not to the move alone.**

**Table 4: Analysis-only EGT fit across all five admitted routes. "Reference" is the $\beta = 2$, $\mu = \beta/Z$ cell; "weak" is the best well-mixed cell with $\beta \leq 0.03$. Values are RMSE in percentage points.**

| Route | Observed shape | Reference | Best weak |
|---|---|---|---|
| Gemini 3 Flash | graded decline | 36.6 | 15.4 |
| GPT-5.4 | graded decline | 32.9 | 5.1 |
| GPT-5.5 | graded decline | 34.9 | 11.3 |
| Claude Sonnet 5 | graded decline | 34.1 | 9.6 |
| Claude Opus 5 | near-step switch | 56.6 | 43.6 |

no matching wheel exists for it, so the sweep ran the pinned pure-Python implementation instead. The quantities above are therefore seeded-chain estimates reported with their between-chain spread, not exact stationary distributions, and the route rates beside them are descriptive prompted self-play endpoints rather than draws from a population process.

*Reading the model backwards.* The sweep above asks what the model predicts. The complementary question is what an observed rate would tell us about the model's own risk parameter, and Figure 4 asks it. The answer bounds how finely this benchmark can be read at all. Because the only place the model bends is its cliff, asking which risk would make it emit an observed rate sends nearly every answer to the same place: the three configured levels span 0.80 of the risk axis, and the 24 invertible route-by-risk cells come back inside a band 0.12 wide, a compression of about sevenfold. What is lost is resolution and not ranking, and the distinction matters. All eight invertible routes return their three implied risks in the configured order, and seven of the eight cells run at 0.1 land strictly left of every cell run at 0.9, the two extreme rows sharing only 0.0006 of the axis. But adjacent rows overlap, the 0.1 and 0.6 rows on 0.0147 and the 0.6 and 0.9 rows on 0.0129, so a rate read back through this model fixes an implied risk only to about a tenth of the axis it was configured on. The inversion runs at $\beta = 0.01$, the weak-selection point, which is the setting that gives the model its best chance of spreading the cells apart; at $\beta = 2$ the inverse is defined only inside a 0.005-wide window, and inverting there would manufacture the compression. Three limits travel with the figure. It is a statement about the model, not an estimate of any route's perceived risk: it says only what the model would have to be handed. A compressed inverse is not an unresponsive route, since all nine routes move the right way with risk and Claude Opus 5 most steeply of all; eight of them merely also land inside the model's own range, which is what lets the inversion answer at all. And every curve behind it is the small-mutation limit, which cannot represent the finite mutation rate of the parameter points it is drawn at.

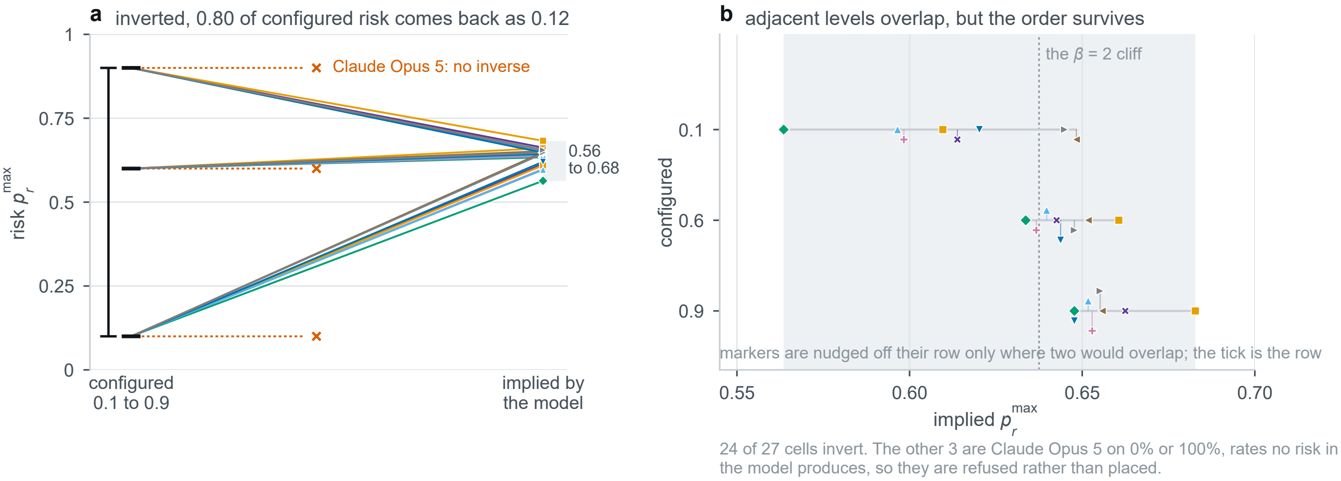



**Figure 4: Inverting the reconstructed evolutionary model through the observed UNSAFE rates of the nine audited routes, at the weak-selection point $\beta = 0.01$. (a) The inversion as a slope chart: configured risk on the left at full scale, the risk the model would need in order to emit the route's observed rate on the right at the same full scale. The fan closes: 0.80 of configured risk comes back as 0.12. Claude Opus 5 has no inverse at any level, because its observed rates of 0% and 100% are rates no risk in the model produces; those three cells are refused rather than placed, and marked as refusals. (b) That band magnified, one row per configured level, with the $\beta = 2$ cliff marked. Adjacent rows overlap, on 0.0147 and 0.0129 of the axis, while the whole configured span of 0.80 collapses to 0.12; what the inversion loses is resolution rather than ranking, since all eight invertible routes keep the configured order and seven of the eight 0.1 cells lie strictly left of every 0.9 cell. Where two markers would fall on each other the second is nudged off its row with a leader back to it; the tick is the row. Neither panel is an estimate of a route's perceived risk, and a compressed inverse is not an unresponsive route.**

## A.7 Crossed representation-robustness design

An earlier local pilot varied the story the game is told through and the symbols used for the two actions, but varied them together with repetition parity, so the two could not be separated, and it failed its comprehension gate. The main paper therefore excludes it and requires a fully crossed rerun before any claim about representation. That rerun has now completed on the two routes selected for this study, and this subsection reports it.

The rerun crosses two *narrative skins*, that is two stories that leave the rules untouched, against two *symbol mappings*, that is which of the two opaque response codes $P$ and $Q$ stands for SAFE and which for UNSAFE, inside repetition blocks, at each of the mechanism's three maximum-private-risk levels. That is twelve cells per route. It completed on both selected routes under protocol `ai-race-frontier-context-mapping-v3`, each with ten repetitions, 120 races, 2,232 recorded decisions and zero parse failures: Gemini 3 Flash as run 1373757 and Claude Sonnet 5 as run 1504455, whose full route strings are given in Table 1. The horizon and setback random-number streams are held common across conditions, so two cells differ in their prompt and not in the environment they met. Table 5 gives all twelve cells for the first route; the second route's cells are in the derived table named at the end of this subsection, and both routes' paired contrasts are in Table 6.

Pooling cells is not the sharpest way to read this design. The repetition index is a common-random-number block, so the same index reuses the same sampled horizon across every skin and mapping cell, and the recorded `horizon_draws_sha256` lets that be checked rather than assumed. It holds for all 60 paired blocks on both routes. Differencing inside a repetition therefore removes the horizon draw from the comparison and leaves the presentation as the only thing that varies. Table 6 reports those paired contrasts, with intervals from a percentile bootstrap over repetition blocks.

**Table 5: Fully crossed narrative-skin by symbol-mapping rerun, first of the two routes that completed it, `google/gemini-3-flash-preview`; the second route, `anthropic/claude-sonnet-5@default`, completed the same twelve cells and its marginals are in the derived table named at the end of this subsection. "Mapping" names which response code stands for SAFE: under `P_SAFE_Q_UNSAFE` the code $P$ means SAFE, and under `Q_SAFE_P_UNSAFE` it means UNSAFE. $n_{race}$ is the number of independent races in the cell and $n_{dec}$ the number of decisions recorded in it; the UNSAFE rate is decision-weighted. All twelve cells are complete and no response failed to parse.**

| Narrative skin | Mapping | Risk | $n_{race}$ | $n_{dec}$ | UNSAFE actions | UNSAFE rate |
|---|---|---|---|---|---|---|
| abstract_game | P_SAFE_Q_UNSAFE | 0.1 | 10 | 186 | 186 | 100.0% |
| abstract_game | P_SAFE_Q_UNSAFE | 0.6 | 10 | 186 | 156 | 83.9% |
| abstract_game | P_SAFE_Q_UNSAFE | 0.9 | 10 | 186 | 111 | 59.7% |
| abstract_game | Q_SAFE_P_UNSAFE | 0.1 | 10 | 186 | 186 | 100.0% |
| abstract_game | Q_SAFE_P_UNSAFE | 0.6 | 10 | 186 | 186 | 100.0% |
| abstract_game | Q_SAFE_P_UNSAFE | 0.9 | 10 | 186 | 166 | 89.2% |
| technology_race | P_SAFE_Q_UNSAFE | 0.1 | 10 | 186 | 186 | 100.0% |
| technology_race | P_SAFE_Q_UNSAFE | 0.6 | 10 | 186 | 144 | 77.4% |
| technology_race | P_SAFE_Q_UNSAFE | 0.9 | 10 | 186 | 81 | 43.5% |
| technology_race | Q_SAFE_P_UNSAFE | 0.1 | 10 | 186 | 186 | 100.0% |
| technology_race | Q_SAFE_P_UNSAFE | 0.6 | 10 | 186 | 185 | 99.5% |
| technology_race | Q_SAFE_P_UNSAFE | 0.9 | 10 | 186 | 83 | 44.6% |

Two things stand out. The symbol mapping moves play by a similar amount on both routes, about eight to ten percentage points, even though the two routes differ by provider and sit six points apart on the admission audit. The narrative skin also moves play on both, but by very different amounts, so skin sensitivity is route-specific in a way that code sensitivity is not. All four intervals exclude zero. Nothing in the mechanism differs across any of these cells: the payoff matrix, the progress increments, the hidden horizon law and the risk formula are identical, and both routes had already passed the comprehension gate. Two routes is enough to show the effect replicates and not enough to characterise endpoints in general.

Panel b of Figure 12 draws both routes' cells as an interaction plot on one common vertical span. That figure is placed with the matched group-size comparison of §A.23, because that is where all three of this supplement's manipulations of the same game can be read side by side. What the panel adds here is the question the table cannot answer: the two factors do not interact on either route, so the narrative skin and the symbol mapping simply add, and the crossing that this rerun was built to test returns an interval spanning zero on both routes.

**Table 6: Repetition-paired presentation contrasts on the two crossed routes. Each contrast is the mean difference in UNSAFE rate within a repetition block, which holds the sampled horizon fixed; positive means the second level raises UNSAFE play. Intervals are 95% percentile bootstrap intervals over the 60 paired blocks. Horizon pairing was verified for 60 of 60 blocks in every row.**

| Route | Contrast | Effect | 95% CI |
|---|---|---|---|
| Gemini 3 Flash | code $Q$ means SAFE | +9.5 pp | +5.6 to +13.8 |
| Claude Sonnet 5 | code $Q$ means SAFE | +8.3 pp | +5.4 to +12.2 |
| Gemini 3 Flash | abstract contest | +10.8 pp | +6.0 to +15.6 |
| Claude Sonnet 5 | abstract contest | +3.4 pp | +0.9 to +7.4 |

Both routes' marginal cells and both paired contrasts are regenerated by `analyze_frontier_context_mapping_cross.py` under `scripts/`, which refuses a route whose race count, ten-per-cell balance or parse-failure count does not check out. Its outputs are `context_mapping_cross_marginals.csv` and a provenance `.json` beside it. An earlier attempt on the Claude route stopped at 106 of 120 races on a provider quota refusal; it is retained under `failed_runs/` as a failure record carrying a `superseded_by` pointer to the complete rerun reported here, run 1504455, and none of its cells is reported, because a counterbalanced design read at 106 of 120 races is unbalanced rather than merely smaller.

Two things are visible in Table 5 and neither is a general finding. First, the risk response survives both manipulations: in every one of the four skin-by-mapping combinations the UNSAFE rate is 100% at risk 0.1 and lower at risk 0.9. Second, the mapping is not inert. Holding the skin fixed, swapping which code means SAFE moves the rate at risk 0.9 from 59.7% to 89.2% under the abstract story and from 43.5% to 44.6% under the technology story, and at risk 0.6 from 83.9% to 100% and from 77.4% to 99.5% respectively. A relabelling that a correct model of the game should ignore therefore changes the action sequence.

What follows from this table is bounded by how many routes stand behind it. The design is crossed on both routes selected for the crossed representation study, Gemini 3 Flash and Claude Sonnet 5, whose full route strings are in Table 1, each at the full twelve cells, so the mapping effect is not confined to one provider's decoder at one revision: it appears on both, at +9.5 and +8.3 points. Two routes are still two routes. They do not separate what belongs to this task from what belongs to commercial endpoints in general, they are not a sample from which a population of models could be described, and the narrative skin already behaves differently on the two, so a third route could as easily break the pattern as extend it. We therefore report the symbol-mapping effect as replicated across the two audited routes, and make no claim about symbol mapping or narrative skin beyond them.

## A.8 Human-benchmark replication check

Before comparing model-generated behaviour with the human benchmark of Domingos and Han [4] in the two-player strategic diversity results in the main paper, we first validated our reconstruction of that benchmark by refitting its published dynamic specification on the same public de-identified data. The reconstructed estimation sample contains 2,888 round-level observations, 172 race-pair clusters, and 338 participants. Table 7 compares the refit coefficients against the values reported in the original study.

Four human sample sizes appear across this supplement, and they differ because each analysis needs a different part of the same public record. The raw first-five-round sequence comparison keeps the 340 participants with a complete five-round action history. The descriptor and elicitation analyses keep 341, because the five aggregate descriptors and the elicited risk score can be computed for one further participant whose complete sequence is unavailable. The dynamic refit above keeps the 338 participants with complete values on every covariate in the published specification. No participant is added anywhere; each figure is a subset of the same released dataset.

**Table 7: Replication of Domingos and Han [4]'s dynamic specification on the public de-identified human data. "Reconstructed" is our refit; "Original" is the coefficient reported in the source study.**

| Term | Reconstructed | Original |
|---|---|---|
| Opponent's previous action | 0.606 | 0.607 |
| Player's progress gap | −0.295 | −0.296 |
| First-round action | 0.217 | 0.217 |
| Player's own previous action | −0.195 | −0.193 |

## A.9 Claim-scoped citation audit

Table 8 records both the supported use of each cited study and the nearest excluded overclaim. These evidence boundaries do not turn the diagnostic pilots in this paper into confirmatory results.

The reconstruction reproduces the original coefficients almost exactly (largest discrepancy 0.002), which is the basis for treating the human-side comparisons in the two-player strategic diversity results in the main paper and the persona and measured risk results

**Table 8: Source-level audit for citations added to the prompt-sensitivity and behavioural-validity argument.**

| Source | Supported use | Excluded overclaim |
|---|---|---|
| Sclar et al. [14] | Meaning-preserving few-shot formatting changes can yield large, model-dependent performance spreads. | Not evidence for semantic paraphrase effects or a universal effect size. |
| Pezeshkpour and Hruschka [11] | Reordering multiple-choice answer options can change predictions. | The proposed uncertainty-plus-placement mechanism is not causally identified. |
| Wei et al. [15] | Order and response-token identity are distinct selection-bias targets. | Not proof that every task or model exhibits the same bias. |
| Salinas and Morstatter [13] | Controlled output-format and atomic whitespace variants can alter answers. | Not a claim that every added space changes behaviour. |
| Zheng et al. [17] | Across their factual-question setup, average persona benefit was absent or slightly negative and reliable persona selection remained difficult. | Not a universal null effect for persona-driven agent tasks. |
| Aher et al. [1] | Three of four behavioural findings were qualitatively reproduced, while the fourth showed a hyper-accuracy distortion. | Not evidence that LLM samples are interchangeable with human populations. |
| Akata et al. [2] | Prediction before action improved coordination and scores in repeated games. | Not a general claim that an observed action reveals a correct opponent belief. |
| Herr et al. [6] | Action-label order and payoff reassignment changed choices in two canonical two-player games. | Not evidence that every strategically equivalent transformation has the same effect. |
| Robinson and Burden [12] | Procedurally varied vignettes produced substantial contextual variability under one underlying Prisoner's Dilemma structure. | Not a repeated-game result and not an estimate for the present race. |
| Wu et al. [16] | Performance declined consistently across 11 tasks when default mappings were replaced by specified counterfactual alternatives. | Not proof that the tested models lacked all abstract reasoning ability. |
| del Rio-Chanona et al. [3] | LLM markets recovered broad feedback-condition dynamics but displayed less strategy heterogeneity than human participants. | Not evidence that LLM agents and human participants are exchangeable. |

in the main paper as resting on a correctly processed benchmark rather than on a reimplementation error.

## A.10 Risk preference and assigned personas

We finally distinguish three constructs that should not be conflated: the human participants' pre-game elicited risk preference, the mechanism's assigned private-risk treatment, and the narrative risk persona supplied to an LLM in its prompt.

The three persona framings themselves are drawn in Figure 5, which the main paper's design section points to rather than reproducing.

Human participants completed an incentivised Eckel–Grossman gamble elicitation (0 = most risk-averse, 5 = least risk-averse) before playing the two-player race: a choice among six gambles, made once, before any round is played. This elicited measure has essentially no direct association with subsequent UNSAFE rate,

$$\text{slope} = -0.004, \qquad r = -0.015, \qquad p = 0.79, \qquad n = 341,$$

where the slope is the trend across the six group means plotted in Figure 6; the participant-level slope of the same relationship is $-0.002$, while the correlation, the $p$ value and the sample size above are participant-level. The difference in mean UNSAFE play between the most and the least risk-accepting human groups, about five percentage points, lies inside the corresponding uncertainty interval. This null result independently reproduces Domingos and Han's own preregistered finding that elicited risk preference did not significantly predict UNSAFE choice [4], and stands in contrast to the tournament literature's general finding that human risk-taking responds to competitive standing rather than remaining a fixed trait [5, 8, 9]: in this task, the round-by-round interaction captured in the two-player strategic diversity results in the main paper is a stronger organising signal than static pre-game disposition. The multiplayer position analysis is exploratory and is not used as a second independent confirmation of this two-player result.

One further place remained where an association could have survived, and it does not. The elicitation might have failed to predict *how much* UNSAFE a participant plays while still predicting *which* dynamic pattern that participant follows. It does not do that either. Comparing the elicited score across the four behavioural archetypes of §A.20, which are the $k$-means groups fitted to five standardised trajectory descriptors of the same human participants, returns a null:

$$H = 4.540, \qquad \text{df} = 3, \qquad p = 0.209, \qquad n = 341,$$

by a tie-corrected Kruskal–Wallis test, which asks whether the distribution of one variable differs across several groups without assuming a normal distribution. The effect size is $\varepsilon^2 = 0.0046$, so the archetype label accounts for under half of one percent of the variance in the elicited score, which is negligible on any convention. No participant is excluded: all 341 human participants carry both an archetype label and an elicited score, and the elicitation is complete.

The archetype medians are identical. All four groups have a median elicited score of 2.0 on the 0 to 5 scale, with interquartile ranges of 1.0 to 2.5 ($n = 19$), 1.0 to 3.0 ($n = 170$), 1.0 to 3.0 ($n = 133$), and 1.5 to 3.5 ($n = 19$). None of the six pairwise comparisons between archetypes is significant after Holm–Bonferroni correction

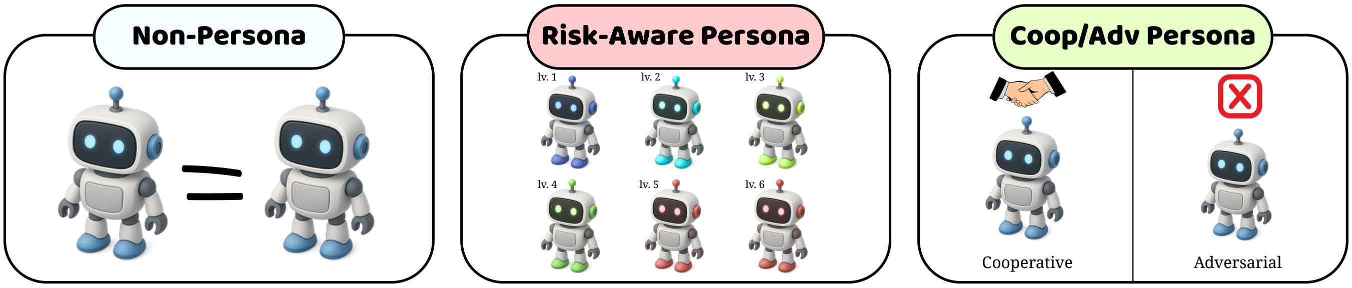



**Figure 5: The persona framings assigned to each company's executive, and the placebo that separates a persona from the mere presence of an extra sentence. Three families are used. The first assigns no persona at all, and is the neutral condition behind every baseline number in this supplement. The second is a risk-aware family with six ordered levels, R1 to R6, whose wording is adapted from the six-choice Eckel–Grossman gamble scale so that R1 is the most risk-averse framing and R6 the least. The third is a social family that frames the rival either as cooperative or as adversarial. The neutral placebo adds a sentence of matched length that carries no risk content and no social content, so a difference between the placebo and no persona at all is an effect of prompt length rather than of persona. All of these are prompt conditions written into the instruction the model receives. None is a trait measured on the agent, and nothing here is elicited from the model, which is the distinction the rest of this section turns on.**

across all six pairs. The smallest uncorrected $p$ is 0.079, between the two largest archetypes, which becomes 0.476 after correction; every other corrected $p$ is above 0.88. The same conclusion follows from Dunn's test on mean ranks and from pairwise Mann–Whitney tests, which the artifact reports side by side.

*Correction.* An earlier version of this supplement reported $H = 21.95$, $p = 0.001$, $n = 286$ for this comparison and read it as a weaker, second-order association surviving the null above. That value has no source anywhere in the analysis record, and no subset of 286 participants arises at any documented step of this pipeline: the elicitation is non-missing for all 341 participants and the clustering drops nobody. It is superseded by the artifact `elicited_risk_by_archetype.json`, whose numbers are the ones reported above.

The null is not a weakness in the comparison; it sharpens it. A measured human risk preference predicts neither the level of UNSAFE play nor the pattern of it, whereas a prompted risk persona moves a checkpoint's UNSAFE rate by between 50.5 and 98.3 percentage points from its lowest to its highest level (Table 9). An elicited disposition and an instructed persona are therefore not two measurements of one construct; they are different kinds of variable, and the shared axis in Figure 6 aligns them for visual comparison only.

Figure 6 (left) shows how little that null leaves. Human mean UNSAFE play stays inside a 49–62% band across the entire elicitation range, never approaching the low-UNSAFE checkpoints of the low-UNSAFE checkpoints of the pilot roster of §A.11 or its high-UNSAFE Gemini checkpoints. Elicited risk preference does not move a participant far enough to resemble a different checkpoint's policy.

The right panel is the opposite (Table 9): every checkpoint with a six-level sweep climbs 50.5 to 98.3 points from R1 to R6, all but the two nano checkpoints monotonically, and those two dip by well under a point at R2 before rising. The label does not, however, move checkpoints by a common factor. GPT-5.6 Terra runs lowest at R1 and finishes within 1.3 points of the top, while GPT-5-nano starts fifth of seven and ends last, so the order of the curves is not

**Table 9: LLM risk-persona effect: change in UNSAFE rate from the lowest (R1) to the highest (R6) assigned risk label, and the per-level OLS slope, for every checkpoint with a six-level risk-label sweep ($n = 360$ players per level, $n = 330$ for Gemini-3-Flash). The persona coefficient (a logistic fit that additionally controls for the mechanism's actual private-risk treatment) has been run only for the first three; "not estimated" marks checkpoints for which only the descriptive change and slope are available. Starred rows sit outside the seven-checkpoint roster.**

| Checkpoint | Δ UNSAFE (pp) | Per-level slope | Coefficient |
|---|---|---|---|
| GPT-5-nano | 50.5 | 0.123 | 0.93 |
| GPT-5.4-nano | 98.2 | 0.212 | 1.47 |
| Gemini-3-Flash | 62.8 | 0.139 | 2.03 |
| Claude Opus 5 | 81.6 | 0.152 | not estimated |
| Claude Sonnet 5 | 61.3 | 0.129 | not estimated |
| GPT-5.6 Luna* | 71.5 | 0.156 | not estimated |
| GPT-5.6 Terra* | 98.3 | 0.203 | not estimated |

preserved, and the same instruction is worth very different amounts to different checkpoints. Of the seven tested checkpoints, five have such a sweep; neither Gemini Flash-Lite checkpoint does. In logistic fits that additionally control for the mechanism's actual private-risk treatment, the coefficients on the agent's own assigned risk label remain large for the three checkpoints where that controlled fit has been run, so the persona effect is not simply standing in for the mechanical risk treatment.

This contrast does not show that LLM agents possess a latent risk preference comparable to the human elicitation: the human variable is a measured pre-game disposition that turns out to be nearly unrelated to in-game behaviour, whereas the LLM variable is an explicit instruction embedded in the prompt. The more defensible reading is that an assigned risk persona functions as a high-salience behavioural instruction for these models, not as an analogue of a

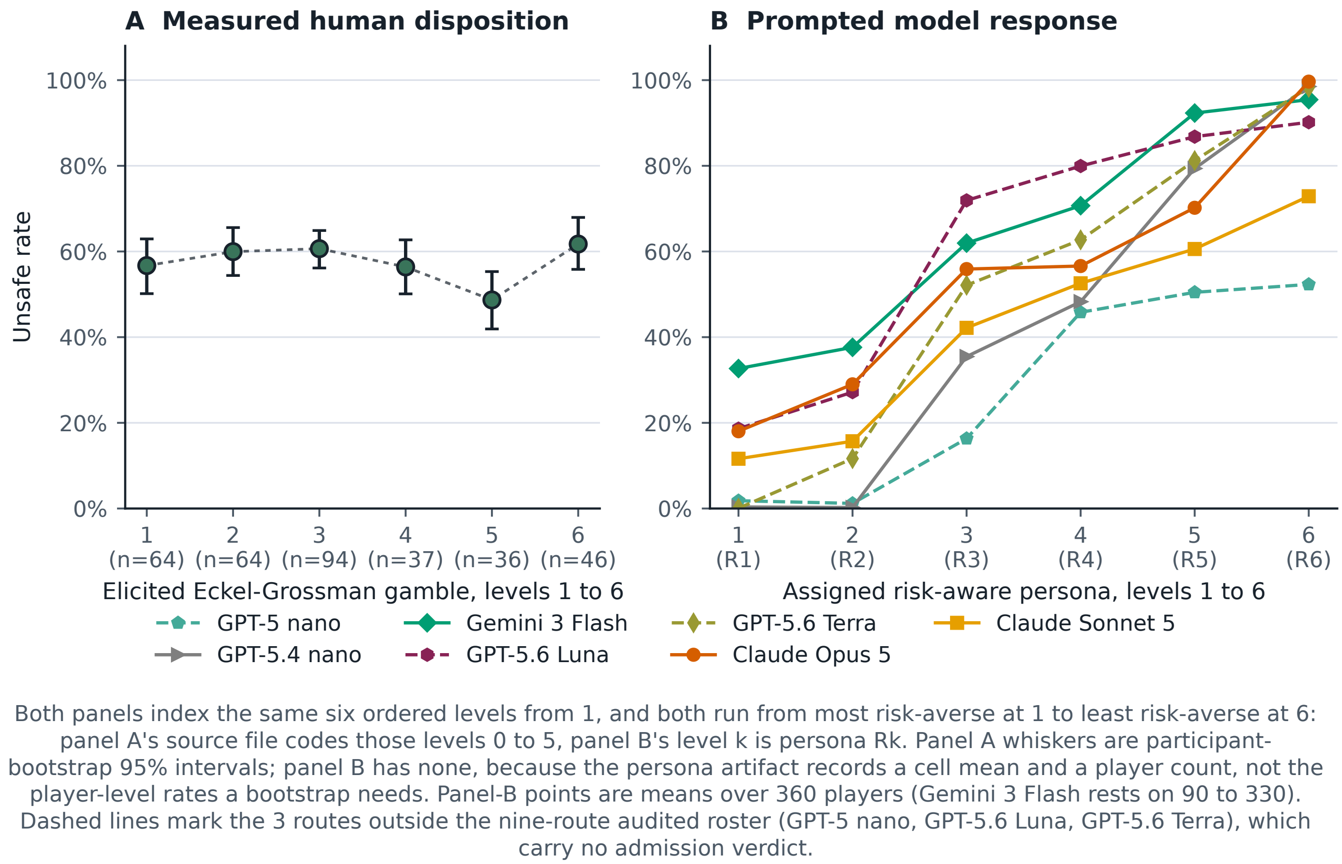



**Figure 6: A measured human disposition against an assigned model label. Left: mean UNSAFE rate by elicited Eckel–Grossman gamble choice, with 95% confidence intervals and participants per bin; the dotted line is the trend across the six group means. Right: mean UNSAFE rate by assigned risk-persona level, for every model with a six-level sweep; GPT-5.6 Luna and Terra (dashed) are the two routes outside the main baseline roster. The shared six-step axis is visual alignment only; the left is a disposition measured once before play, the right an instruction re-assigned every race.**

stable human risk disposition. Because the LLM estimates come from a separate risk-matrix experiment, available for five of the seven tested checkpoints and with the private-risk-controlled logistic fit run for only three of those five, this comparison remains exploratory and cross-experimental rather than a like-for-like measurement.

### A.11 The pilot roster behind the projections

The diversity comparison in the main paper reads all nine audited routes on the confirmatory baseline. The projections in this appendix, the behavioural archetypes and the two-dimensional embedding, are built on an earlier export of the same two-player game covering seven checkpoints, and that roster is described here so those projections can be read without it occupying main-paper space. It pools 420 trajectories, 60 per checkpoint, with 340 complete human trajectories from the public de-identified dataset of Domingos and Han [4], one incomplete human record excluded, $N = 760$. Six of the seven checkpoints carry an endpoint-admission verdict from §A.1; GPT-5 nano is no longer offered on the audited account and so cannot be gated at all, and is retained as description rather than as evidence. Each player is represented by 15 raw features: own action, opponent action, and own-minus-opponent progress gap entering each of rounds 1 to 5, which are the recorded states and actions themselves rather than derived quantities. The archetype projection uses 341 participants rather than 340, because its five descriptor summaries can be computed for one further participant whose complete five-round sequence is unavailable.

No single aggregate rate characterises these checkpoints. Mean UNSAFE play over rounds 1 to 5 is lowest for GPT-5 nano (17%), then Claude Sonnet 5 (22%), Claude Opus 5 (33%), GPT-5.4 nano (54%) and humans (56%), with the Gemini checkpoints running from 73% (Gemini 3 Flash) to 83% (Gemini 3.1 Flash Lite). Closeness to the human mean is, on its own, uninformative about anything beyond that one number, which is why the main paper measures diversity on the action sequences instead.

The archetype projection places each trajectory at the nearest of four behavioural archetypes fitted to the human sample alone

**Table 10: Race-grouped population-identity diagnostic on 760 complete first-five-round trajectories. The eight populations are imbalanced (340 human and 60 per checkpoint), so balanced accuracy and macro-F1 are primary. The null interval is the 2.5–97.5 percentile range over grouped label permutations.**

| Metric | Observed | Permutation null mean | Null 95% interval |
|---|---|---|---|
| Raw accuracy | $0.292 \pm 0.053$ | 0.086 | 0.039–0.178 |
| Balanced accuracy | $\mathbf{0.423 \pm 0.043}$ | 0.124 | 0.070–0.182 |
| Macro-F1 | $\mathbf{0.252 \pm 0.047}$ | 0.063 | 0.025–0.112 |

(§A.22). The human sample occupies every archetype at every cluster count from two to six, while each checkpoint except GPT-5.4 nano is more concentrated, so that conclusion does not depend on the choice of four (§A.20); the archetypes themselves are exploratory descriptions rather than recovered types, best separated at four groups and only moderately stable under bootstrap. The two-dimensional embedding of the same space is §A.16, and neither the main paper nor this appendix rests a conclusion on it: recolouring its identical coordinates by outcome rather than by population separates it almost entirely by UNSAFE rate.

The heterogeneity those pictures suggest is also testable directly, and the test belongs here because it covers five checkpoints rather than the nine the main paper reads. Fitting three nested logistic models to the pooled neutral-baseline pilot, risk only, checkpoint identity only, and checkpoint identity by risk, shows that letting the risk-response profile vary by checkpoint improves fit substantially beyond checkpoint-specific intercepts alone, $\chi^2(10) = 354.7$, $p < 10^{-3}$, with several checkpoint-by-risk cells approaching complete separation. A convergence warning in the interaction model means the displayed likelihood-ratio statistic is approximate, so only the conservative threshold is quoted. The fit covers the five-checkpoint pilot baseline only, and the two Claude checkpoints are not in it. Neither caveat changes what it supports: no single risk-response curve represents all five, so the differences between checkpoints are differences in *shape* and not only in level.

## A.12 Population identity in first-five-round trajectories

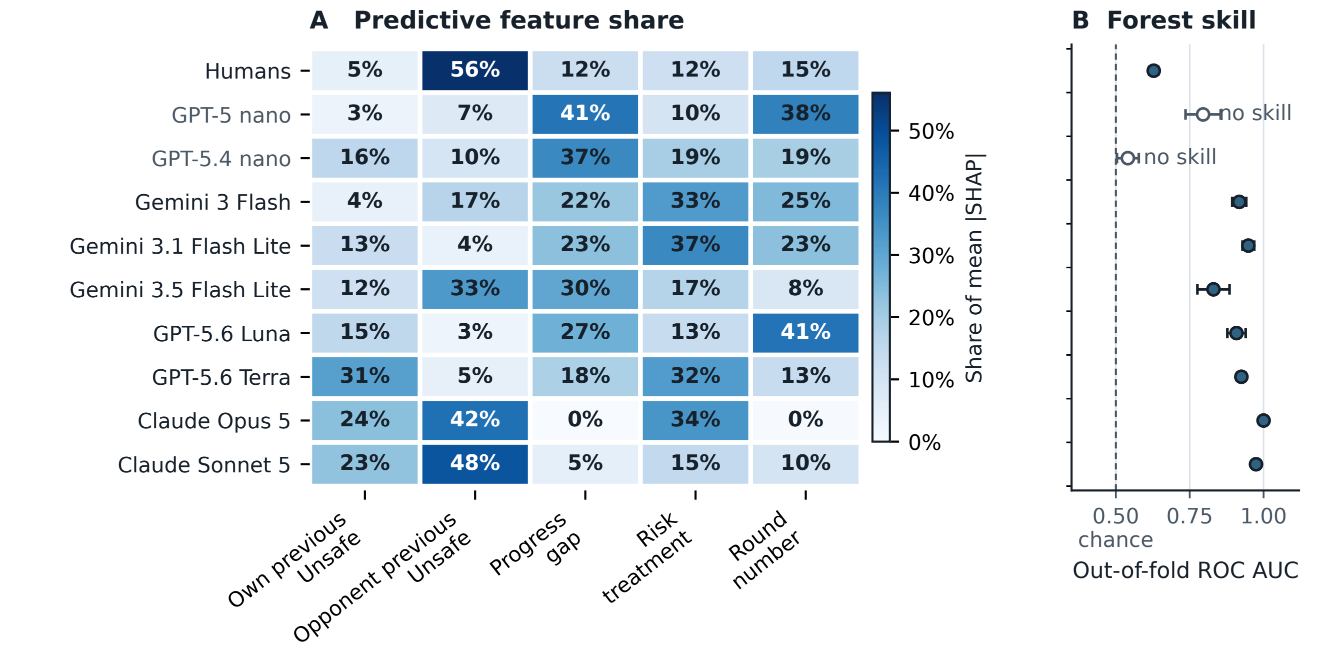
Table 10 is the diagnostic the main paper summarises: it asks how well a simple classifier can name the population that produced a trajectory, using the same 15 raw features as the trajectory embedding of §A.16. We fit a shallow decision tree with class-balanced loss under stratified five-fold grouped cross-validation. The two seats of an LLM race remain in the same fold; human observations are grouped by participant. Since the pooled classes are unequal, balanced accuracy and macro-F1 are the primary measures. A 200-repetition permutation null preserves the observed class counts and the same folds.

The balanced accuracy exceeds the class-balanced chance level of 1/8 and the grouped permutation null ($p < 0.001$); raw accuracy is shown only for completeness because it is affected by the large human class. This result supports observable differences in the first-five-round trajectories, while the overlap and the shallow classifier do not support a claim that a latent strategy or model identity is cleanly recoverable. The machine-readable result and source hashes are stored in `results/cross_model_pilot_synthesis/data/`.

## A.13 Predictive structure of UNSAFE choices

The preceding section shows that populations produce different action patterns. We now ask which recorded parts of the game best predict their next UNSAFE choice. This is a study of association, not cause or internal reasoning. For each population, we fit a random forest, a model that combines many decision trees. Table 11 reports the result and Figure 7 shows the same shares as a heatmap. Predictive scores use five-fold grouped cross-validation, with all decisions from one participant or race kept in one fold. The forest predicts decisions after round one from five values known before the choice: the player's previous action, the opponent's previous action, the progress gap, the assigned maximum private risk, and the round number. TreeSHAP then measures how much each value contributes to the fitted model's predictions. A larger SHAP share means that the predictor relied more on that feature. It does not mean that the feature caused the decision.

For human participants, the opponent's previous action accounts for 56% of total SHAP magnitude, far more than any other feature, while the player's own previous action contributes only 5%: a reciprocity-dominated profile that independently recovers the pattern found in the human logistic analysis. One caveat has to travel with that reference profile everywhere it is used below, because it is the yardstick the checkpoints are held against and it is the least well identified row in the table. The human forest barely discriminates: ROC AUC 0.63, balanced accuracy 0.58, and a test accuracy of 0.618 against a majority-class baseline of 0.588, so it is 3.0 points of accuracy above predicting the more common action and nothing else. Adding the demographic covariates does not help, AUC 0.62. The model-side forests are far better fitted, up to AUC 1.00 for Claude Opus 5 and 0.97 for Claude Sonnet 5, so every comparison in this subsection is between shares extracted from very differently identified models. That is not a reason to drop the analysis, and it is a reason not to read a matched ordering as a matched policy. GPT-5-nano is close to the opposite profile: relative race position (progress gap) accounts for 41% of its predictive importance and opponent history for only 7%, and its apparently strong out-of-fold AUC of 0.80 is misleading on its own, since balanced accuracy is exactly 0.50: the checkpoint plays so close to the SAFE floor that the classifier effectively fails to recover the rare UNSAFE class. Gemini-3-Flash and Gemini-3.1-Flash-Lite are primarily risk-treatment-driven (33% and 37% of SHAP magnitude), consistent with their strong, monotone descriptive risk-response curves. Claude Sonnet 5 is the checkpoint closest to the human profile, and it is the exception to any claim that no model shares it: the opponent's previous action is its largest predictor too, at 48% against the human 56%. Be exact about how far that goes. Two other checkpoints also rank that feature first, Claude Opus 5 at 42%, which the dagger above marks as collinear with the risk treatment rather than a reciprocity estimate, and Gemini-3.5-Flash-Lite at 33% with its progress gap close behind at 30%. And beyond the leading feature no checkpoint reproduces the human rank order: below the opponent term the human profile runs round number,




Associational feature use by separately fitted forests; it is not a causal attribution. Panel A normalises every row to 100% whether or not the forest predicts anything, so panel B carries the skill each row rests on: whiskers are the cross-validated standard deviation, and an open marker flags a forest whose balanced accuracy is at chance and whose accuracy does not beat the majority class.

**Figure 7: Population-specific predictive structure for round-$t \geq 2$ UNSAFE choices. (A) Each row reports the share of mean absolute SHAP value assigned to the same five pre-decision features by a random forest fitted separately to that population. SHAP magnitudes describe how the fitted classifier uses observed features; they are neither causal effects nor evidence of the agents' internal reasoning. (B) The out-of-fold discrimination each row rests on, because panel A normalises every row to 100% whether or not its forest predicts anything; whiskers are the cross-validated standard deviation and an open marker flags a forest at chance. Read the two panels together, because they are not on the same footing. The human forest discriminates weakly, ROC AUC 0.63 and balanced accuracy 0.58, only 3.0 points of accuracy above guessing the majority class, while the two Claude forests reach 1.00 and 0.97. A share from a weakly fitted forest says mostly where that forest found the little signal there was, so it is a much looser proxy for the player's policy than a share from a well fitted one.**

**Table 11: Population-specific random-forest prediction of UNSAFE play. Feature columns report shares of total mean absolute SHAP magnitude (rows sum to 100%, up to rounding); AUC and balanced accuracy are mean out-of-fold values from the population-specific race/participant-grouped five-fold split.**

| Population | Own previous | Opponent previous | Progress gap | Risk | Round | OOF AUC | Balanced accuracy |
|---|---|---|---|---|---|---|---|
| Human | 5% | **56%** | 12% | 12% | 15% | 0.63 | 0.58 |
| GPT-5-nano | 3% | 7% | **41%** | 10% | 38% | 0.80 | 0.50 |
| GPT-5.4-nano | 16% | 10% | **37%** | 19% | 19% | 0.54 | 0.52 |
| Gemini-3-Flash | 4% | 17% | 22% | **33%** | 25% | 0.92 | 0.77 |
| Gemini-3.1-Flash-Lite | 13% | 4% | 23% | **37%** | 23% | 0.95 | 0.75 |
| Gemini-3.5-Flash-Lite | 12% | **33%** | 30% | 17% | 8% | 0.83 | 0.71 |
| Claude Opus 5 | 24% | 42%† | 0% | 34% | 0% | 1.00 | 1.00 |
| Claude Sonnet 5 | 23% | **48%** | 5% | 15% | 10% | 0.97 | 0.93 |

†Collinear with the risk treatment, not a reciprocity estimate; see the discussion below.

progress gap, risk, own previous, while Claude Sonnet 5 runs own previous, risk, round number, progress gap. The agreement is in which feature leads, not in the ordering as a whole. GPT-5.4-nano has no comparably dominant predictor: importance is spread across progress gap, risk, round, and its own previous action, with opponent history at 10% and the lowest above-chance predictive fit in the table (AUC 0.54, balanced accuracy 0.52), so its behaviour is comparatively difficult to reconstruct from these five mechanical state variables rather than clearly governed by any one signal.

Claude Opus 5 shows why these shares must be read against the design that produced them. Its 42% opponent share reads as strong reciprocity and is not: on the neutral lane it plays UNSAFE on essentially every low-risk decision and essentially none at high risk, so the risk treatment alone separates its choices, and since both seats behave that way the opponent's previous action is a proxy for the risk level rather than an independent signal. The forest then splits importance between two variables encoding the same event, and the perfect AUC describes a step function, not a better fit. A feature-importance share is interpretable only where the feature keeps variation independent of the others, which is what a near-deterministic policy removes.

One marginal association does agree across populations, and it is the part of this analysis that the main paper's diversity section points at: human winners play UNSAFE 16.0 percentage points more often than human losers, against an approximately 20-point pooled difference among the checkpoints. The dynamic structure behind that agreement does not agree. GPT-5.4-nano in particular has no single strong predictor, and several checkpoint estimates are weak, reversed in sign, or not reliably estimable because their action distributions sit at the floor or the ceiling.

Predictive performance varies substantially across populations (AUC 0.54 to 1.00), so SHAP shares should not be read as directly comparable measures of decision quality; floor and ceiling behaviour mechanically limits minority-class prediction for several models. The defensible conclusion is narrower than "LLMs use different features": the same observable state variables organise predictive information differently across populations, and matching a human aggregate action rate does not imply matching the human reciprocity-dominated association profile.

## A.14 Two-player checkpoint descriptive baseline (full scope)

The five-checkpoint two-player descriptive baseline referenced throughout the two-player strategic diversity results in the main paper and the observable drivers of unsafe play in the main paper completed 150 races and 2,790 decisions, crossing five checkpoints (GPT-5-nano, GPT-5.4-nano, Gemini-3-Flash-preview, Gemini-3.1-Flash-Lite, Gemini-3.5-Flash-Lite) with the mechanism's three assigned private-risk levels ($0.10, 0.60, 0.90$; Eq. (2)). At the level of descriptive curves rather than the pooled aggregate rates already reported in the main text, GPT-5-nano's UNSAFE rate was low and nearly flat across the three risk levels; GPT-5.4-nano's response to risk level was non-monotone; and the three Gemini checkpoints' UNSAFE rate declined as the assigned maximum risk increased, the opposite direction from the two OpenAI checkpoints. Because provider routes, protocol signatures, and pilot sample sizes differ across checkpoints, these five curves demonstrate qualitative heterogeneity in how each checkpoint responds to the risk manipulation; we do not pool them into a single model-family estimate, and the formal test of that heterogeneity is the checkpoint-by-risk interaction of §A.11. Table 12 gives the per-checkpoint, per-risk-level rates underlying the qualitative curve descriptions above.

**Table 12: Player-level UNSAFE rate by checkpoint and assigned maximum-private-risk level, five-checkpoint two-player descriptive baseline ($n$ = 20 players per checkpoint–risk cell, ten races each).**

| Checkpoint | Risk 0.10 | Risk 0.60 | Risk 0.90 |
|---|---|---|---|
| GPT-5 nano | 12.3% | 15.1% | 14.5% |
| GPT-5.4 nano | 57.7% | 49.6% | 56.7% |
| Gemini 3 Flash | 100.0% | 72.3% | 53.9% |
| Gemini 3.1 Flash Lite | 100.0% | 80.1% | 69.9% |
| Gemini 3.5 Flash Lite | 83.8% | 70.8% | 62.6% |

## A.15 $N$-player self-play scope ($N = 3, 4, 5$)

The exploratory multiplayer position analysis reports position effects for GPT-5-nano and GPT-5.4-nano at fixed $N$; this subsection gives the broader group-size scope. Matched OpenAI self-play baselines cover $N \in \{3, 4, 5\}$ for both checkpoints: 180 pilot races, 720 player trajectories, and 6,120 decisions in total, pooled across the mechanism's three risk levels. Table 13 reports the aggregate UNSAFE rate by checkpoint and group size.

**Table 13: Aggregate UNSAFE rate by group size, OpenAI self-play, pooled across the three assigned private-risk levels. Ten independent races per model–risk–$N$ cell.**

| Checkpoint | $N = 3$ | $N = 4$ | $N = 5$ |
|---|---|---|---|
| GPT-5-nano | 16.2% | 26.8% | 21.9% |
| GPT-5.4-nano | 83.7% | 87.0% | 84.0% |

Neither checkpoint's response to group size is monotone in $N$, and the association is checkpoint-specific: GPT-5-nano's rate roughly doubles from $N = 3$ to $N = 4$ before partially receding at $N = 5$, while GPT-5.4-nano stays within a narrow 83–87% band throughout. We do not read either pattern as an isolated group-size effect, for two reasons. First, changing $N$ also changes the stage-payoff rule itself (Eq. (3)–(4) use the joint count $k^t$ of SAFE-choosing companies rather than a single opponent's move), the number of opponents, and prompt length, so $N$ is confounded with the representation of the game as well as with its size. Second, the pilots use only ten independent races per model–risk–$N$ cell, which is a small base for the per-cell rates entering Table 13 and leaves individual cells vulnerable to the same kind of small-$n$ noise flagged for specific rank × risk-band cells in the exploratory position analysis. Table 14 disaggregates Table 13 by assigned risk level and reports a 95% percentile bootstrap interval over the ten independent races in each cell. The decision count is retained only as a descriptive row count.

## A.16 Trajectory embedding recoloured by outcome

The embedding of raw five-round trajectories separates almost entirely by how much UNSAFE play a trajectory carries. Both colourings of it live here. Figure 8 colours it by population, and Figure 9 is the check: the same coordinates, with only the colouring changed, so any structure visible in the second is structure the embedding

**Table 14: UNSAFE rate by checkpoint, group size, and assigned maximum-private-risk level, OpenAI $N$-player self-play. Intervals are 95% percentile bootstrap intervals over race clusters, and $n_{\text{race}}$ is the independent-unit count.**

| $N$ | Checkpoint | Risk | $n_{\text{race}}$ | UNSAFE rate (95% CI) |
|---|---|---|---|---|
| 3 | GPT-5 nano | 0.10 | 10 | 17.4% (14.5–20.0%) |
| 3 | GPT-5 nano | 0.60 | 10 | 18.1% (13.3–22.4%) |
| 3 | GPT-5 nano | 0.90 | 10 | 11.3% (6.1–16.9%) |
| 3 | GPT-5.4 nano | 0.10 | 10 | 85.4% (82.5–88.5%) |
| 3 | GPT-5.4 nano | 0.60 | 10 | 83.7% (78.5–88.7%) |
| 3 | GPT-5.4 nano | 0.90 | 10 | 83.3% (80.0–86.9%) |
| 4 | GPT-5 nano | 0.10 | 10 | 19.2% (14.5–23.8%) |
| 4 | GPT-5 nano | 0.60 | 10 | 27.8% (21.6–33.3%) |
| 4 | GPT-5 nano | 0.90 | 10 | 24.1% (15.5–32.1%) |
| 4 | GPT-5.4 nano | 0.10 | 10 | 88.1% (85.9–90.3%) |
| 4 | GPT-5.4 nano | 0.60 | 10 | 88.4% (86.2–90.6%) |
| 4 | GPT-5.4 nano | 0.90 | 10 | 83.8% (80.3–87.5%) |
| 5 | GPT-5 nano | 0.10 | 10 | 18.9% (14.7–22.4%) |
| 5 | GPT-5 nano | 0.60 | 10 | 20.0% (16.1–23.6%) |
| 5 | GPT-5 nano | 0.90 | 10 | 22.7% (18.7–26.9%) |
| 5 | GPT-5.4 nano | 0.10 | 10 | 80.0% (75.4–84.9%) |
| 5 | GPT-5.4 nano | 0.60 | 10 | 87.1% (84.5–89.6%) |
| 5 | GPT-5.4 nano | 0.90 | 10 | 85.1% (82.5–88.0%) |

already had. Reading them in that order is the point. The population panels look like eight different regions; recolouring the identical points by each player's own UNSAFE rate splits the embedding into a SAFE region, an UNSAFE region and a thin mixed band, with very few points on the wrong side. The apparent separation of populations is therefore largely a separation of action rates, which is why the main paper rests no conclusion on either picture and uses the grouped classifier of Table 10 instead.

## A.17 Mean round-by-round profile per population

The embedding in the main paper shows that human trajectories spread more widely than any single checkpoint's. Figure 10 is the same comparison read one feature at a time: population means over the fourteen features every trajectory shares, with action probabilities and absolute progress gaps separated so that neither scale obscures the other.

## A.18 Player-level UNSAFE rate by population

Figure 11 is the aggregate level of the same comparison: one number per population, the mean share of a player's first five decisions that were UNSAFE. It is the plainest form of the point that a matching aggregate rate does not imply matching behaviour, because a population can reach a middling rate either by mixing or by every player behaving the same way.

## A.19 UNSAFE rate by persona band and relative position

The main paper does not report these cells. Tables 15 and 16 give them in full for the two checkpoints that have a persona sweep and an observed position. Relative position is observed rather than assigned, so these are associations within a persona band, not position effects. The race is this paper's independent unit, but the per-race source for these two specific breakdowns could not be located in the analysis record. We therefore print only the decision count and descriptive rate, with no confidence interval or inferential interpretation.

**Table 15: UNSAFE rate by persona band and relative position, GPT-5-nano. $n$ is the number of decisions in the cell. These cells are descriptive only and support no inference.**

| Persona band | Position | $n$ | UNSAFE rate |
|---|---|---|---|
| Baseline | Leader | 1460 | 19.7% |
| Baseline | Middle | 1065 | 30.4% |
| Baseline | Trailer | 175 | 33.1% |
| Low (R1–R2) | Leader | 1094 | 5.9% |
| Low (R1–R2) | Middle | 198 | 14.6% |
| Low (R1–R2) | Trailer | 58 | 12.1% |
| Mid (R3–R4) | Leader | 649 | 40.7% |
| Mid (R3–R4) | Middle | 451 | 38.1% |
| Mid (R3–R4) | Trailer | 250 | 32.8% |
| High (R5–R6) | Leader | 702 | 63.2% |
| High (R5–R6) | Middle | 339 | 54.3% |
| High (R5–R6) | Trailer | 309 | 50.2% |

**Table 16: UNSAFE rate by persona band and relative position, GPT-5.4-nano. $n$ is the number of decisions in the cell and cells marked † have $n < 20$. These cells are descriptive only and support no inference.**

| Persona band | Position | $n$ | UNSAFE rate |
|---|---|---|---|
| Baseline | Leader | 1649 | 83.6% |
| Baseline | Middle | 703 | 84.8% |
| Baseline | Trailer | 348 | 86.5% |
| Low (R1–R2) | Leader | 1073 | 3.3% |
| Low (R1–R2) | Middle | 268 | 1.5% |
| Low (R1–R2) | Trailer | 9† | 0.0% |
| Mid (R3–R4) | Leader | 1164 | 95.7% |
| Mid (R3–R4) | Middle | 70 | 88.6% |
| Mid (R3–R4) | Trailer | 116 | 91.4% |
| High (R5–R6) | Leader | 1328 | 99.3% |
| High (R5–R6) | Middle | 4† | 100.0% |
| High (R5–R6) | Trailer | 18† | 100.0% |

## A.20 Cluster-count sensitivity and stability of the human archetypes

The behavioural archetypes used in this supplement and in the main paper are $k$-means groups: the 341 human participants are described by five standardised summary numbers each, and $k$-means partitions them into $k$ groups so that each participant is near the average of its own group. The number of groups is a choice, not a measurement, so this subsection reports what changes when that choice changes. The five descriptors are the overall UNSAFE rate, reciprocity, position sensitivity, own-action autocorrelation, and whether the first round was UNSAFE. They are standardised on the

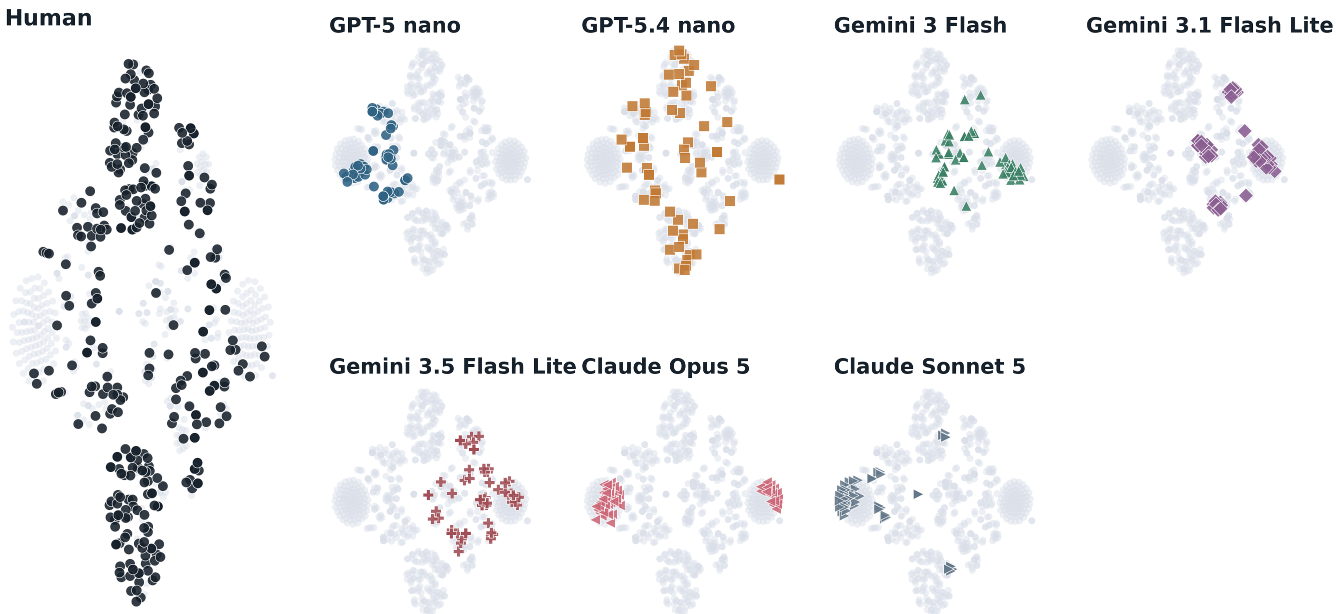



**Figure 8: The population-coloured embedding, shown here so that Figure 9 can be compared against it directly. A t-SNE embedding places each of the 760 complete first-five-round trajectories at a point in two dimensions, arranged so that trajectories with similar action-and-position histories fall near each other; the input is the pooled 15-dimensional first-five-round action-and-position space, the same input as the population-identity classifier below. One panel is drawn per population, human first. The axes have no units and no direction: only relative closeness carries meaning, and distances between well-separated clusters are not interpretable. The main paper rests no conclusion on this figure, and the reason is Figure 9: recolouring these identical coordinates by each player's own UNSAFE rate separates the embedding almost entirely, so what looks like a separation of populations here is largely a separation of action rates. The quantitative statement about population identity is the grouped classifier of Table 10, not this picture.**

human sample alone, and every other population is later projected into that same human space rather than re-standardised.

Table 17 gives the diagnostics for $k = 2$ to 6. Three of them describe how tight and separated the groups are: the silhouette coefficient, which is higher when a participant is closer to its own group than to the nearest other one; the Calinski–Harabasz index, higher when between-group spread is large relative to within-group spread; and the Davies–Bouldin index, which is *lower* when groups are better separated. The fourth, inertia, is the total squared distance to group centres and falls mechanically as $k$ grows, so it is reported for completeness rather than as evidence. The choice of $k = 4$ used elsewhere is the best cell on two of the three interpretable indices, with the highest silhouette (0.316, against 0.278 at $k = 2$ and 0.241 at $k = 5$) and the lowest Davies–Bouldin (1.218).

Stability is only moderate, and we say so rather than presenting the groups as recovered types. The bootstrap adjusted Rand index falls from 0.961 at $k = 2$, where the split is nearly always the same one, to 0.717 at $k = 3$, 0.626 at $k = 4$, 0.458 at $k = 5$, and 0.518 at $k = 6$. At $k = 4$ the interval runs from 0.39 to 0.98, which is wide. The groups are therefore useful descriptions of where the human sample sits in this five-number space, not evidence that four discrete human strategies exist.

The conclusion the main paper draws from these groups does not depend on $k$. That conclusion is about coverage: the human sample spreads across the whole set of archetypes while almost every single checkpoint concentrates on a few. Table 18 counts, for each population and each $k$, how many of the $k$ archetypes that population actually occupies, where a population counts as occupying an archetype when at least 5% of its entities project to it. The human row occupies all $k$ archetypes at every $k$ from two to six. No checkpoint matches that. The closest is the exception the main paper names, GPT-5.4 nano, which occupies all $k$ archetypes at $k = 2$, 3, 5 and 6 and three of four at $k = 4$. Every other checkpoint falls short from $k = 4$ upward, and GPT-5 nano is the extreme case, occupying exactly one archetype at every $k$. The 5% threshold is a reporting choice, and the artifact records the counts at a 1% threshold and at strictly positive occupancy alongside it so the claim can be read at any of the three.

One property of these descriptors is ours to disclose rather than a reviewer's to discover. Two of the five are undefined for some participants, and the clustering fills the gap with the sample mean instead of dropping the participant, exactly as the published clustering does. Position sensitivity, which contrasts a participant's behaviour when ahead against when behind, is undefined for 268 of the 341 participants, because those participants were never observed both ahead and behind; it is mean-imputed for all 268, that is for 79% of the sample. Reciprocity and own-action autocorrelation

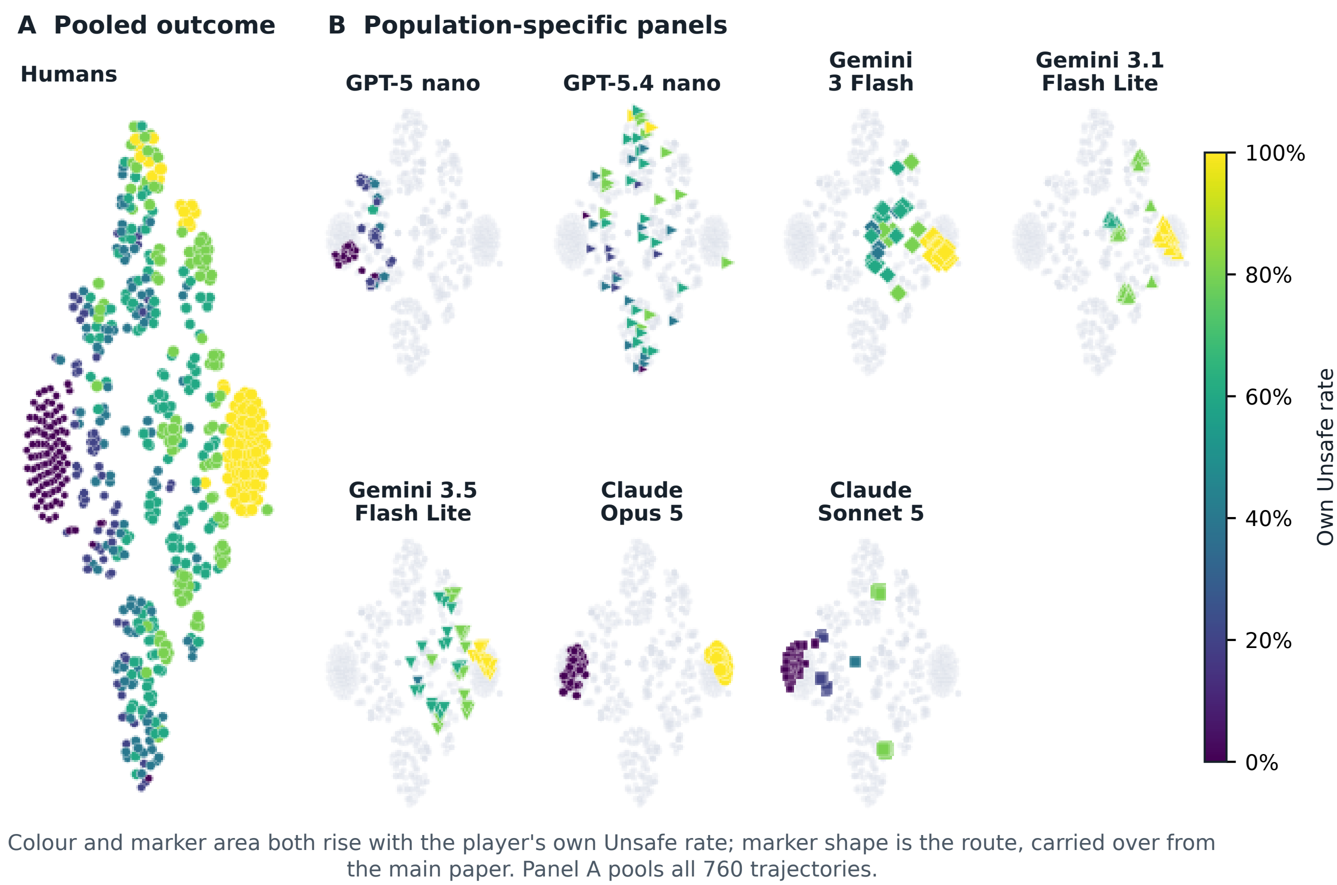



**Figure 9: The same t-SNE coordinates as Figure 8, recoloured by each player's own mean UNSAFE rate over rounds 1–5 (green = SAFE, red = UNSAFE) instead of population identity. Left: all 760 trajectories pooled. Right: the same colouring broken out per population.**

**Table 17: Cluster-count sweep for the human behavioural archetypes, $k = 2$ to $6$, on the same matrix of 341 human participants by five standardised descriptors, seed 0. Higher is better for silhouette and Calinski–Harabasz, lower is better for Davies–Bouldin; inertia falls with $k$ by construction. "Group sizes" is the number of participants per group. The adjusted Rand index (ARI) measures how much a partition refitted on a resampled sample agrees with the partition fitted on the full sample: 1 is identical, 0 is chance agreement. It is computed over 500 bootstrap resamples of participants, and the interval is the 2.5 to 97.5 percentile range across those resamples.**

| $k$ | Silhouette | Calinski Harabasz | Davies Bouldin | Inertia | Group sizes | Smallest group $n$ | ARI mean | ARI 95% interval |
|---|---|---|---|---|---|---|---|---|
| 2 | 0.278 | 110.9 | 1.578 | 1280.9 | 187 / 154 | 154 | 0.961 | 0.76–1.00 |
| 3 | 0.267 | 91.7 | 1.542 | 1102.1 | 107 / 63 / 171 | 63 | 0.717 | 0.37–0.98 |
| 4 | 0.316 | 83.5 | 1.218 | 975.3 | 19 / 170 / 133 / 19 | 19 | 0.626 | 0.39–0.98 |
| 5 | 0.241 | 78.2 | 1.363 | 880.2 | 20 / 57 / 118 / 87 / 59 | 20 | 0.458 | 0.30–0.69 |
| 6 | 0.239 | 79.0 | 1.405 | 780.2 | 42 / 78 / 56 / 55 / 91 / 19 | 19 | 0.518 | 0.34–0.79 |

are each undefined and mean-imputed for 38 participants. The overall UNSAFE rate and the first-round action are complete for all 341. A mean-imputed cell carries no information about the participant it is attached to, so position sensitivity does most of its work on under a quarter of the sample, and the archetype signatures should be read with that in mind. No participant is excluded at any step: all 341 receive a cluster label.

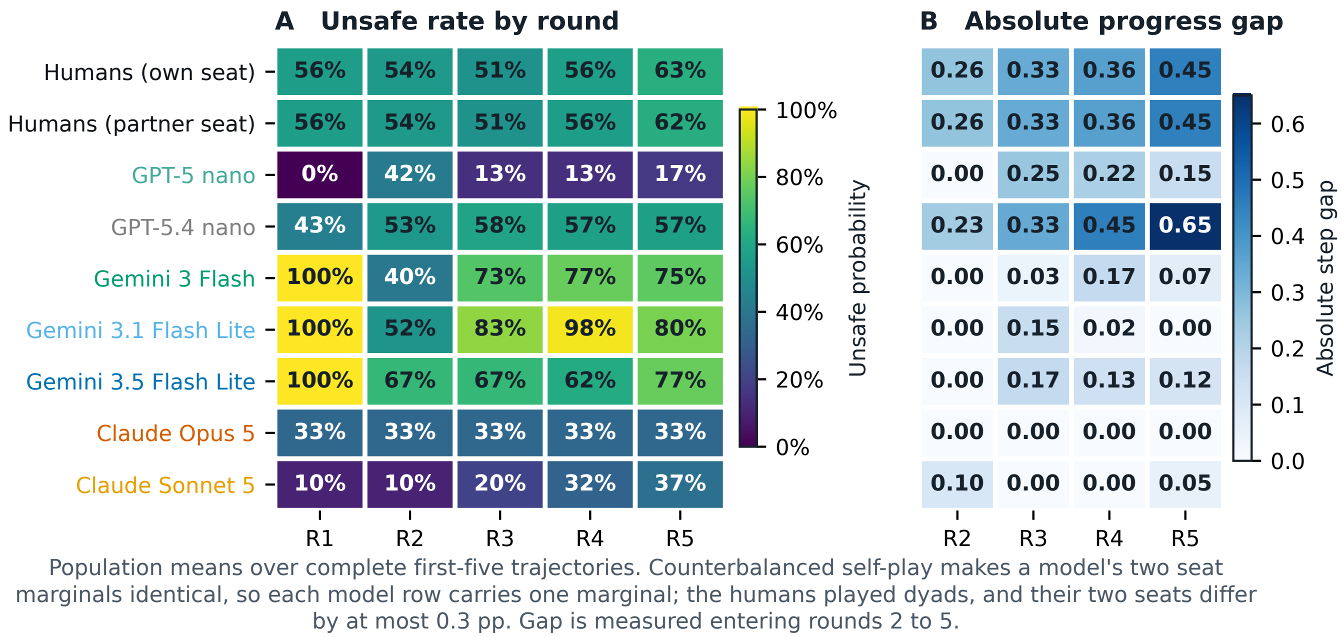



**Figure 10: Round-by-round profile for each population, shown as two source-backed heatmaps. Panel A reports the probability of an UNSAFE action for each player's own and opponent action at rounds 1–5. Panel B reports the mean absolute progress gap entering rounds 2–5, so it measures how far apart the two players are after pooling focal-player roles. The eight rows are the human reference and the seven baseline model populations.**

**Table 18: Archetype occupancy by population and cluster count. Each cell counts how many of the $k$ human-fitted archetypes that population occupies, where occupying means that at least 5% of the population's entities project nearest to that archetype. $n$ is the number of entities projected: participants for the human row, no-persona player-races for each checkpoint. A row that reaches $k$ at every $k$ covers the full human-fitted space; the human reference does so at every $k$ and no checkpoint does.**

| Population | $n$ | $k$=2 | $k$=3 | $k$=4 | $k$=5 | $k$=6 |
|---|---|---|---|---|---|---|
| Human (reference) | 341 | 2 | 3 | 4 | 5 | 6 |
| GPT-5 nano | 120 | 1 | 1 | 1 | 1 | 1 |
| GPT-5.4 nano | 120 | 2 | 3 | 3 | 5 | 6 |
| Gemini 3 Flash | 120 | 1 | 1 | 2 | 3 | 3 |
| Gemini 3.1 Flash Lite | 60 | 1 | 1 | 2 | 2 | 3 |
| Gemini 3.5 Flash Lite | 60 | 1 | 1 | 2 | 3 | 3 |
| GPT-5.6 Luna | 120 | 2 | 1 | 2 | 2 | 3 |
| GPT-5.6 Terra | 120 | 2 | 2 | 3 | 3 | 4 |
| Claude Opus 5 | 120 | 2 | 2 | 2 | 2 | 2 |
| Claude Sonnet 5 | 120 | 2 | 3 | 2 | 3 | 4 |

## A.21 Trajectory-diversity statistics

The main paper reports that human first-five-round trajectories are more varied than those of every route the gate admits. Two tables give the numbers behind that, one row per population per risk level, so a single value can be checked against its population, its risk condition, and the number of independent units it rests on. Table 20 is the one the main paper's claim rests on: all nine audited routes, on the confirmatory baseline. Table 19 is the earlier seven-checkpoint pilot export, kept because it carries GPT-5 nano, an endpoint no longer offered on the audited identity and therefore holding no admission verdict. That export has no row for GPT-5.4 or GPT-5.5, which are two of the five admitted routes, so it cannot settle a claim about the admitted set and is not used for one.

A trajectory here is the pair of actions taken by the two players over the first five rounds, that is ten SAFE or UNSAFE bits. Three statistics describe how varied a set of such trajectories is. Hill $q = 0$ counts how many *distinct* trajectories appear, so it treats a trajectory seen once and a trajectory seen forty times alike. Hill $q = 1$ is the effective number of trajectories, the exponential of Shannon entropy, which discounts trajectories that are rare: it equals $q = 0$ when every trajectory is equally common and falls below it when a few dominate. Mean pairwise Hamming distance is the average number of the ten bits on which two trajectories drawn from the set differ, divided by ten, so 0 means every trajectory in the set is identical and 0.5 means two trajectories differ on half their bits on average.

Sample sizes differ across populations, and both counting statistics rise with sample size, so every cell is compared at the same size. The point estimate is the mean over draws of 20 trajectories taken without replacement from the cell, or the exact cell value where the cell already holds exactly 20. The interval is a two-stage cluster bootstrap: clusters are resampled with replacement first, and only then is the draw rarefied to 20 inside each resample, over 2,000 resamples. The cluster is the independent unit of the design:

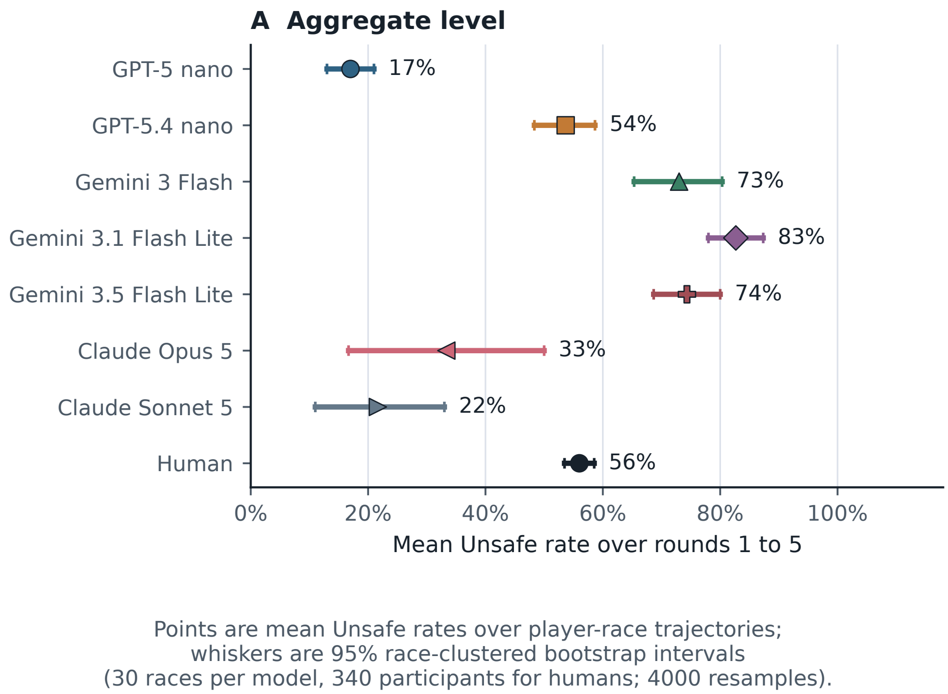



**Figure 11: Player-level mean UNSAFE rate over rounds 1 to 5, one row per population. The point is the unweighted mean over player-race trajectories. The whisker is a 95% percentile bootstrap interval that resamples *clusters*, not single trajectories, because the two seats of a race are not independent of each other: they share the sampled stopping horizon and the setback draws. The cluster is therefore the race for every checkpoint row, 30 races each, and the participant for the human row, 340 participants; the interval uses 4,000 resamples. The point estimates are unchanged from a trajectory-level calculation; only the interval changed, and it supersedes an earlier interval that resampled player-race trajectories and so treated the two seats of a race as independent.**

a participant for the human rows, one per trajectory, and a race for the checkpoint rows, because the two seats of a race share the same sampled horizon and setback draws and so are not independent of each other. The $n_{\text{cluster}}$ column is that count.

The ordering in Table 19 is consistent across all three risk levels and all three statistics. The human rows sit near the ceiling of the comparison: 19.2, 19.7, and 19.5 distinct trajectories out of a possible 20, with mean pairwise Hamming distance between 0.467 and 0.497, close to the 0.5 that two unrelated trajectories would give. Only one checkpoint in this pilot export reaches that: GPT-5.4 nano returns 19.0 distinct trajectories at every risk level, with Hamming between 0.478 and 0.506. The rest are far below. Claude Opus 5 returns exactly one distinct trajectory at all three risk levels, with Hamming 0, meaning every one of its player-races produced the identical ten-action sequence. Gemini 3 Flash and Gemini 3.1 Flash Lite collapse the same way at risk 0.1. Claude Sonnet 5 falls from 5.0 distinct trajectories at risk 0.1 to 3.0 at risk 0.9, with Hamming 0.020 at the latter. This is the shape of the finding: a checkpoint can match a human aggregate action rate while producing a small handful of distinct action sequences, or only one.

Table 20 is the version of that comparison that can be read against the admission gate, because it has a row for every route the gate ruled on. Each of the five admitted routes sits entirely below the human Hill $q$=1 interval at all three risk levels; the largest admitted upper bound anywhere in the table is 12.3, at risk 0.9, where the human lower bound is 14.8. Two routes do not sit entirely below, GPT-5.4 mini and GPT-5.4 nano, and neither passed the gate: they score 75.0% and 51.7%, the latter the lowest of the nine. Claude Opus 5 again returns exactly one distinct trajectory at all three risk levels, with Hamming 0. The direction of the finding therefore does not depend on which of the two tables is read. What the confirmatory table adds is that the claim about the admitted set can be checked over the whole admitted set, which the pilot export could not do.

## A.22 Behavioural-archetype coverage by population

The main paper reports that human trajectories spread more evenly across the archetypes than most single checkpoints do, with GPT-5.4-nano the exception. Table 21 is the per-population breakdown behind that statement at the four archetypes used throughout, so a single share can be checked against its population and its archetype; Table 18 is the same comparison read as a count, at every cluster count from two to six.

**Table 19: Sample-size-matched diversity of paired first-five-round trajectories, by maximum-private-risk level and population. A trajectory is the two players' ten actions over rounds 1 to 5. Hill $q$=0 is the number of distinct trajectories, Hill $q$=1 the effective number after discounting rare ones, and Hamming the mean fraction of the ten bits on which two trajectories differ. $n_{\text{src}}$ is the trajectories available in the cell and $n_{\text{cluster}}$ the independent units it rests on: participants for the human rows, races for the checkpoint rows. Every cell is compared at 20 trajectories. The parenthesised ranges are 95% *resampling intervals* (RI) and not confidence intervals: they are two-stage cluster bootstrap ranges, with clusters resampled with replacement and then rarefied to 20 within each resample, over 2,000 resamples, and they bound how much the statistic moves under resampling rather than making a confidence statement about the observed count. Because the point estimate is the observed cell value while the interval is computed on resampled clusters, which lose distinct trajectories to duplication, an interval can sit below its point estimate.**

| Risk | Population | $n_{\text{src}}$ | Hill $q$=0 (95% RI) | Hill $q$=1 (95% RI) | Hamming (95% RI) | $n_{\text{cluster}}$ |
|---|---|---|---|---|---|---|
| 0.1 | Human | 98 | 19.2 (15.0–20.0) | 18.9 (13.2–20.0) | 0.467 (0.407–0.504) | 98 |
| | GPT-5 nano | 20 | 13.0 (5.0–12.0) | 10.2 (3.4–10.7) | 0.275 (0.163–0.339) | 10 |
| | GPT-5.4 nano | 20 | 19.0 (9.0–16.0) | 18.7 (7.3–15.2) | 0.478 (0.405–0.505) | 10 |
| | Gemini 3 Flash | 20 | 1.0 (1.0–1.0) | 1.0 (1.0–1.0) | 0.000 (0.000–0.000) | 10 |
| | Gemini 3.1 Flash Lite | 20 | 1.0 (1.0–1.0) | 1.0 (1.0–1.0) | 0.000 (0.000–0.000) | 10 |
| | Gemini 3.5 Flash Lite | 20 | 5.0 (3.0–5.0) | 3.6 (1.9–4.8) | 0.166 (0.076–0.227) | 10 |
| | Claude Opus 5 | 20 | 1.0 (1.0–1.0) | 1.0 (1.0–1.0) | 0.000 (0.000–0.000) | 10 |
| | Claude Sonnet 5 | 20 | 5.0 (3.0–5.0) | 4.5 (2.6–5.0) | 0.354 (0.246–0.413) | 10 |
| 0.6 | Human | 104 | 19.7 (15.0–20.0) | 19.7 (14.1–20.0) | 0.495 (0.455–0.513) | 104 |
| | GPT-5 nano | 20 | 8.0 (5.0–8.0) | 7.2 (3.4–7.7) | 0.244 (0.154–0.301) | 10 |
| | GPT-5.4 nano | 20 | 19.0 (9.0–16.0) | 18.7 (7.2–15.2) | 0.506 (0.402–0.520) | 10 |
| | Gemini 3 Flash | 20 | 11.0 (5.0–11.0) | 9.3 (4.4–10.2) | 0.312 (0.208–0.371) | 10 |
| | Gemini 3.1 Flash Lite | 20 | 4.0 (2.0–4.0) | 2.8 (2.0–3.7) | 0.229 (0.211–0.255) | 10 |
| | Gemini 3.5 Flash Lite | 20 | 12.0 (6.0–12.0) | 11.5 (5.5–10.9) | 0.364 (0.267–0.397) | 10 |
| | Claude Opus 5 | 20 | 1.0 (1.0–1.0) | 1.0 (1.0–1.0) | 0.000 (0.000–0.000) | 10 |
| | Claude Sonnet 5 | 20 | 5.0 (1.0–5.0) | 2.7 (1.0–4.0) | 0.074 (0.000–0.133) | 10 |
| 0.9 | Human | 138 | 19.5 (16.0–20.0) | 19.3 (14.8–20.0) | 0.497 (0.457–0.516) | 138 |
| | GPT-5 nano | 20 | 12.0 (6.0–12.0) | 10.0 (3.9–10.7) | 0.232 (0.134–0.304) | 10 |
| | GPT-5.4 nano | 20 | 19.0 (9.0–16.0) | 18.7 (7.3–15.2) | 0.486 (0.394–0.504) | 10 |
| | Gemini 3 Flash | 20 | 11.0 (6.0–11.0) | 10.7 (5.5–10.2) | 0.311 (0.262–0.314) | 10 |
| | Gemini 3.1 Flash Lite | 20 | 3.0 (2.0–3.0) | 2.9 (2.0–3.0) | 0.101 (0.079–0.105) | 10 |
| | Gemini 3.5 Flash Lite | 20 | 12.0 (6.0–12.0) | 11.5 (5.2–10.9) | 0.401 (0.319–0.417) | 10 |
| | Claude Opus 5 | 20 | 1.0 (1.0–1.0) | 1.0 (1.0–1.0) | 0.000 (0.000–0.000) | 10 |
| | Claude Sonnet 5 | 20 | 3.0 (1.0–3.0) | 1.5 (1.0–2.3) | 0.020 (0.000–0.054) | 10 |

## A.23 Matched group-size comparison

The three- to five-player pilots reported elsewhere in this supplement have no two-player arm, so a difference between group sizes could not be told apart from a difference between two separately built lanes. This subsection reports a matched comparison that removes that particular ambiguity.

Every group size, including two, is run through the same $N$-player engine, the same prompt template, the same parser and one protocol, `ai-race-nplayer-matched-hosted-confirmatory-v1`, so the number of seats is the only thing deliberately varied. The two-player arm is not a near neighbour of the headline game: evaluating the group-count stage rule of the main paper at two companies returns its two-player payoff matrix exactly, at 1.0, 0.6, 2.4, 2.0, which `scripts/verify_matched_nplayer_design.py` asserts offline before the task is pushed, together with the invariance of every other mechanism field and the sharing of the horizon draw described below.

*What was collected, and how.* The full grid: every group size at every private-risk condition, $p_r^{\max} \in \{0.1, 0.6, 0.9\}$, ten repetitions each, on the admitted route `google/gemini-3-flash-preview`. That is 120 races and 3,696 decisions with zero parse failures. No Model Proxy identity available to us sustains the whole four-by-three grid, which needs roughly 7,800 requests, so the sweep was partitioned into the analysis unit itself, one (group size, risk) cell of ten repetitions, and each cell was collected whole on one identity so that it carries its own race-clustered interval rather than being stitched together across accounts. Cells were assigned to identities in a plan committed before the first was started, and the assignment rotates across risk levels so that no identity is confounded with a group size; every cell records its collecting identity in the artifact. Every cell's bootstrap generator is derived from a digest of that cell alone, so collecting one condition cannot move an interval already reported for another, and `scripts/analyze_nplayer_matched.py` refuses to report any risk level that is missing an arm.

Each cell was collected on a different Kaggle identity, again by a plan fixed in advance rather than by retrying until something succeeded: $N = 2$ on one account, $N = 3$, $N = 4$ and $N = 5$ on three others, each cell collected whole so that it carries its own race-clustered interval. The identity is a billing boundary rather than a model boundary, since the route, prompt, parser, protocol, temperature request and seed structure are identical across cells, but it is a difference between runs and is recorded per cell in the artifact rather than argued away.

*The contrast is paired on the horizon.* The game seed is base + rep and does not depend on the treatment name or on the number of seats, so one repetition index reuses a single horizon stopping-draw stream at every group size. Differencing a group size against the

**Table 20: The same comparison on the confirmatory baseline, which holds all nine audited routes at ten races per risk level and so at the same matched size of 20. Columns, statistics, rarefaction and the 95% resampling interval (RI), which is not a confidence interval, are exactly those of Table 19; only the source of the trajectories differs, and the main paper's admission verdict is printed beside each route. Every admitted route sits entirely below the human Hill $q$=1 interval at all three risk levels. The two routes that do not, GPT-5.4 mini and GPT-5.4 nano, are both routes the gate refused.**

| Risk | Population | $n_{\text{src}}$ | Hill $q$=0 (95% RI) | Hill $q$=1 (95% RI) | Hamming (95% RI) | $n_{\text{cluster}}$ |
|---|---|---|---|---|---|---|
| 0.1 | Human | 98 | 19.2 (15.0–20.0) | 18.9 (13.2–20.0) | 0.467 (0.407–0.504) | 98 |
| | Gemini 3 Flash (admitted) | 20 | 3.0 (1.0–3.0) | 1.5 (1.0–2.3) | 0.020 (0.000–0.054) | 10 |
| | Claude Opus 5 (admitted) | 20 | 1.0 (1.0–1.0) | 1.0 (1.0–1.0) | 0.000 (0.000–0.000) | 10 |
| | GPT-5.4 (admitted) | 20 | 5.0 (3.0–5.0) | 4.0 (1.9–4.9) | 0.171 (0.076–0.234) | 10 |
| | GPT-5.5 (admitted) | 20 | 11.0 (3.0–11.0) | 6.3 (2.2–8.9) | 0.145 (0.058–0.216) | 10 |
| | Claude Sonnet 5 (admitted) | 20 | 2.0 (2.0–2.0) | 2.0 (2.0–2.0) | 0.211 (0.211–0.211) | 10 |
| | Gemini 3.1 Flash Lite (refused) | 20 | 3.0 (1.0–3.0) | 1.5 (1.0–2.3) | 0.020 (0.000–0.054) | 10 |
| | GPT-5.4 mini (refused) | 20 | 20.0 (10.0–16.0) | 20.0 (8.3–15.2) | 0.416 (0.330–0.458) | 10 |
| | Gemini 3.5 Flash Lite (refused) | 20 | 10.0 (6.0–10.0) | 10.0 (5.0–9.5) | 0.282 (0.200–0.331) | 10 |
| | GPT-5.4 nano (refused) | 20 | 16.0 (7.0–14.0) | 15.2 (5.8–13.2) | 0.418 (0.294–0.446) | 10 |
| 0.6 | Human | 104 | 19.7 (16.0–20.0) | 19.7 (14.4–20.0) | 0.495 (0.456–0.514) | 104 |
| | Gemini 3 Flash (admitted) | 20 | 14.0 (6.0–13.0) | 12.3 (4.9–11.7) | 0.333 (0.251–0.363) | 10 |
| | Claude Opus 5 (admitted) | 20 | 1.0 (1.0–1.0) | 1.0 (1.0–1.0) | 0.000 (0.000–0.000) | 10 |
| | GPT-5.4 (admitted) | 20 | 11.0 (5.0–11.0) | 8.4 (3.4–10.0) | 0.257 (0.145–0.297) | 10 |
| | GPT-5.5 (admitted) | 20 | 13.0 (5.0–13.0) | 10.7 (4.2–11.5) | 0.326 (0.259–0.352) | 10 |
| | Claude Sonnet 5 (admitted) | 20 | 5.0 (4.0–5.0) | 4.9 (3.0–5.0) | 0.180 (0.132–0.205) | 10 |
| | Gemini 3.1 Flash Lite (refused) | 20 | 7.0 (3.0–7.0) | 5.8 (3.0–6.8) | 0.186 (0.094–0.254) | 10 |
| | GPT-5.4 mini (refused) | 20 | 17.0 (8.0–15.0) | 16.2 (6.6–14.1) | 0.360 (0.251–0.426) | 10 |
| | Gemini 3.5 Flash Lite (refused) | 20 | 13.0 (7.0–13.0) | 11.5 (4.8–11.7) | 0.380 (0.290–0.411) | 10 |
| | GPT-5.4 nano (refused) | 20 | 19.0 (9.0–16.0) | 18.7 (7.3–15.2) | 0.459 (0.358–0.481) | 10 |
| 0.9 | Human | 138 | 19.5 (16.0–20.0) | 19.3 (14.8–20.0) | 0.497 (0.461–0.516) | 138 |
| | Gemini 3 Flash (admitted) | 20 | 12.0 (5.0–12.0) | 10.0 (4.0–10.7) | 0.414 (0.349–0.419) | 10 |
| | Claude Opus 5 (admitted) | 20 | 1.0 (1.0–1.0) | 1.0 (1.0–1.0) | 0.000 (0.000–0.000) | 10 |
| | GPT-5.4 (admitted) | 20 | 14.0 (6.0–13.0) | 12.5 (5.0–12.3) | 0.414 (0.251–0.480) | 10 |
| | GPT-5.5 (admitted) | 20 | 13.0 (5.0–12.0) | 10.7 (3.9–10.7) | 0.269 (0.155–0.349) | 10 |
| | Claude Sonnet 5 (admitted) | 20 | 7.0 (5.0–7.0) | 6.0 (3.2–6.8) | 0.267 (0.152–0.331) | 10 |
| | Gemini 3.1 Flash Lite (refused) | 20 | 7.0 (5.0–7.0) | 6.4 (3.4–6.8) | 0.305 (0.194–0.371) | 10 |
| | GPT-5.4 mini (refused) | 20 | 17.0 (7.0–15.0) | 15.2 (5.5–14.1) | 0.408 (0.258–0.477) | 10 |
| | Gemini 3.5 Flash Lite (refused) | 20 | 13.0 (7.0–13.0) | 12.3 (5.6–11.7) | 0.403 (0.316–0.417) | 10 |
| | GPT-5.4 nano (refused) | 20 | 16.0 (8.0–14.0) | 15.2 (6.3–13.2) | 0.412 (0.267–0.465) | 10 |

**Table 21: Behavioural-archetype coverage: share of each population's neutral-lane player-races assigned to each human-fit archetype. Human row is the reference clustering ($n$=341); each LLM row projects that checkpoint's own no-persona player-races ($n$=60 to 120) into the same space. This table draws on the full nine-checkpoint cross-model pilot and therefore includes GPT-5.6 Luna and Terra, which are not part of the seven-checkpoint trajectory comparison in the main paper.**

| Population | Cautious | Aggr./recip. | Catch-up | Persister |
|---|---|---|---|---|
| Human (reference) | 39.0% | 49.9% | 5.6% | 5.6% |
| GPT-5-nano | 99.2% | 0% | 0% | 0.8% |
| GPT-5.4-nano | 51.7% | 33.3% | 3.3% | 11.7% |
| Gemini-3-Flash | 1.7% | 78.3% | 20.0% | 0% |
| Gemini-3.1-Flash-Lite | 0% | 83.3% | 16.7% | 0% |
| Gemini-3.5-Flash-Lite | 0% | 76.7% | 23.3% | 0% |
| GPT-5.6 Luna | 3.3% | 72.5% | 24.2% | 0% |
| GPT-5.6 Terra | 14.2% | 59.2% | 26.7% | 0% |
| Claude Opus 5 | 66.7% | 33.3% | 0% | 0% |
| Claude Sonnet 5 | 77.5% | 20.8% | 1.7% | 0% |

two-player arm inside a repetition therefore removes the horizon draw from the comparison. The analyser re-derives that pairing from the recorded game seed instead of assuming it, and it holds for all ten blocks in every contrast.

*What it shows.* UNSAFE play rises monotonically with the number of competing companies at every private-risk condition that leaves room for it to rise, and all six of those contrasts exclude zero. Because the two-player arm is the headline game itself, this also says something about the rest of the paper: the two-player results are the low end of that curve, not a special case beside it.

Figure 12 draws that grid as its first panel, beside the two other manipulations this supplement ran on the same game: the crossed representation design of §A.7 and the seat asymmetry of the neutral baseline. The figure sits in this subsection rather than with the representation design for two reasons. Group size is the only one of the three factors that moves play a long way, so it is the panel the figure leads with; and by this point in the supplement the other two designs have already been described in full, whereas a reader meeting the figure at §A.7 would meet this grid many pages before the subsection that defines it. Read as a surface rather than as a table, the grid also makes plain something the rows do not. Adding companies does not carry play up a plane. It carries play into the ceiling, and the five hatched cells are where the measurement stops being a measurement: those cells contained no safe decision at all, so their contrast against the two-player arm is bounded by the boundary and not by the route.

The size of the effect is what makes the grid worth having. Going from two companies to three adds 11.1 points at $p_r^{\max} = 0.6$ and 11.0 at 0.9; going to four adds 28.2 and 28.1. The two risk conditions were

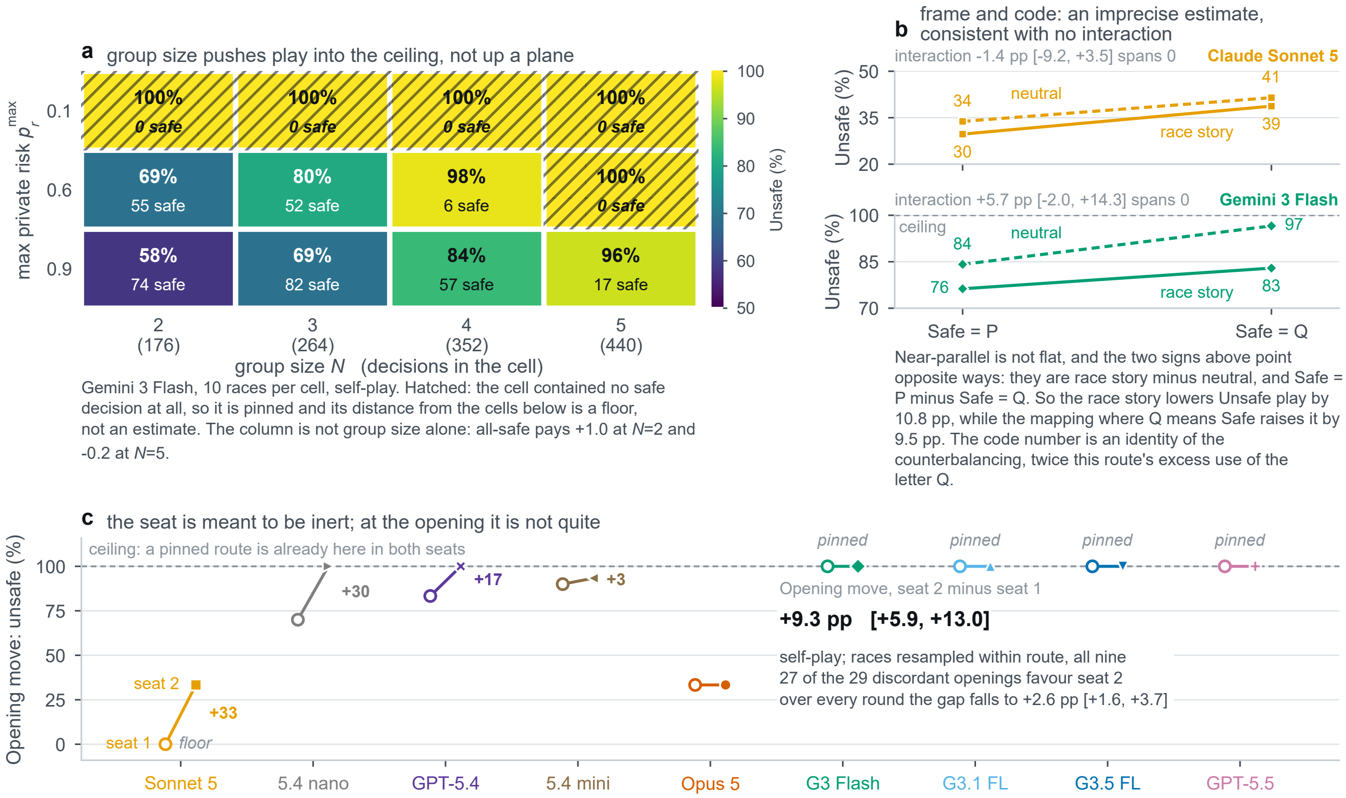



**Figure 12: Three manipulations on the same game, and only one of them moves play a long way. (a) The matched group-size grid of Table 22 drawn as a surface, on `google/gemini-3-flash-preview`, ten races per cell and 3,696 decisions in all. Colour is the UNSAFE rate; the second line in each cell is the number of safe decisions the cell actually contained. Five of the twelve cells contained none, and those are hatched, because a cell with no safe decision left in it has no room to move and its distance from the cells below is a floor rather than an estimate. The column is also not group size alone: playing safe while everyone else does pays +1.0 at $N = 2$ and $-0.2$ at $N = 5$. (b) The crossed representation design of §A.7 on both routes selected for the crossed representation study, drawn on a common thirty-point vertical span so that a slope in the upper panel is the same slope as in the lower one. The lines run close to parallel and the interaction spans zero on both routes, $-1.4$ pp $[-9.2, +3.5]$ on Claude Sonnet 5 and $+5.7$ pp $[-2.0, +14.3]$ on Gemini 3 Flash, which is an imprecise estimate consistent with no interaction rather than a demonstration that there is none. Near-parallel is not flat, and both main effects are present on both routes: the race story lowers UNSAFE play by 3.4 and 10.8 points against the neutral frame, and the mapping under which $Q$ means SAFE raises it by 8.3 and 9.5 points. The two contrasts are signed in opposite physical directions, which the figure now states in words; the code contrast is also an algebraic identity of the counterbalancing, exactly twice the route's excess use of the letter $Q$, so it says which response surface carries the effect and does not on its own explain it. The letter use is no fixed habit: Gemini 3 Flash emits $Q$ on 77% of its turns when $Q$ means UNSAFE and on 11% when it means SAFE. (c) The two seats, whose prompts are symmetric word for word, at the opening move of all 270 neutral-baseline races. Four routes open UNSAFE from both seats in all thirty of their races and sit on the marked ceiling, where a seat effect could not appear at all; one seat opened UNSAFE in none of its races and is marked at the floor. Pooled over all nine routes, with races resampled inside their own route, the opening gap is +9.3 pp [+5.9, +13.0], and over every round of the race it falls to +2.6 pp [+1.6, +3.7]. Three limits travel with the figure. Panel a is one route, so it does not separate a property of group size from a property of this route in groups. An interaction that spans zero is inconclusive rather than absent, the more so on the route whose interval reaches +14 points, so panel b is not evidence that representation is inert. And because four routes in panel c are at the ceiling in both seats and can only pull the average toward zero, its pooled gap is a floor on the seat asymmetry rather than a measure of its size. No interval here is corrected for asking three questions on one figure.**

collected on different days and different identities, and they agree to a tenth of a point. The five-company contrast is the exception, +31.0 against +40.0, and the reason is visible in the table rather than interesting: at $p_r^{max} = 0.6$ the five-company cell is already at 100%, so its contrast is truncated by the boundary and understates

**Table 22: Matched group-size comparison on `google/gemini-3-flash-preview`. Ten races per cell, zero parse failures throughout. Rates carry 95% percentile bootstrap intervals over races; the contrast is the mean difference against the two-player arm within a repetition block, with its interval taken over the ten blocks. Horizon pairing was verified for 10 of 10 blocks in every row. At $p_r^{\max} = 0.1$ every cell is at the ceiling, so its intervals have zero width: that is an absence of variation in the sample, not precision. The five-company cell at $p_r^{\max} = 0.6$ is at the ceiling too, so its contrast there is truncated, which leaves the high-risk condition as the only one that measures the five-company effect against no boundary.**

| $N$ | Decisions | UNSAFE rate (95% CI) | minus $N$=2 (95% CI) | |
|---|---|---|---|---|
| *Low private risk, $p_r^{\max} = 0.1$* | | | | |
| 2 | 176 | 100.0% (100.0–100.0) | baseline | |
| 3 | 264 | 100.0% (100.0–100.0) | +0.0 pp | (+0.0 to +0.0) |
| 4 | 352 | 100.0% (100.0–100.0) | +0.0 pp | (+0.0 to +0.0) |
| 5 | 440 | 100.0% (100.0–100.0) | +0.0 pp | (+0.0 to +0.0) |
| *Medium private risk, $p_r^{\max} = 0.6$* | | | | |
| 2 | 176 | 68.8% (65.2–72.4) | baseline | |
| 3 | 264 | 80.3% (76.0–83.8) | +11.1 pp | (+6.3 to +15.1) |
| 4 | 352 | 98.3% (95.6–100.0) | +28.2 pp | (+22.6 to +33.2) |
| 5 | 440 | 100.0% (100.0–100.0) | +31.0 pp | (+26.4 to +35.3) |
| *High private risk, $p_r^{\max} = 0.9$* | | | | |
| 2 | 176 | 58.0% (50.7–63.7) | baseline | |
| 3 | 264 | 68.9% (62.3–74.7) | +11.0 pp | (+5.9 to +16.4) |
| 4 | 352 | 83.8% (80.6–86.2) | +28.1 pp | (+24.5 to +32.0) |
| 5 | 440 | 96.1% (92.3–98.6) | +40.0 pp | (+33.2 to +47.4) |

the effect. The high-risk condition is the only one that measures the five-company step against no ceiling.

The low risk condition is where the comparison becomes informative about its own limits. There every cell is at 100%, so the group-size contrast is exactly zero at every size. The right reading is not that group size stops mattering when private risk is low. It is that this route is already at the ceiling before any competitor is added, so the design has no room left in which an effect could appear. The same route reaches 98.9% and 100.0% UNSAFE at $p_r^{\max} = 0.1$ in the two independent neutral-baseline runs of §A.5, collected under a different protocol on a different engine, so the ceiling is a property of the route at low risk rather than an artifact of this sweep. Read together with the ceiling at $p_r^{\max} = 0.6$ for five companies, the pattern is that this measurement has power only where play is off the boundary, which is a statement about the instrument and not about where the mechanism operates.

*What it does not show.* Group size cannot be separated from everything that comes with it, and no analysis repairs that. The group-count rule makes the stage payoff depend on how many companies there are, and the state description names each of them, so a larger group is also a different payoff and a longer prompt. The estimand is therefore the effect of group size *within this game's own definition.* The comparison covers one route and ten races per cell, so it is not a claim about endpoints or about a population of models. It is regenerated by `scripts/analyze_nplayer_matched.py`, which refuses to report a risk level unless all four group sizes are present, refuses a cell with a parse failure or the wrong seat count, and records the collecting identity of every cell it uses.

*Which route carries the seat gap.* Panel c of Figure 12 reports a gap averaged over nine routes, and a reader is entitled to ask whether it describes nine routes or one. `scripts/analyze_seat_confound.py` answers that from the same corpus and writes to `results/derived/seat_confound/`. Over every round of the race the pooled gap of +2.6 pp is carried almost entirely by GPT-5.4 nano, at +19.5 pp [+12.9, +25.7]. Set that one route aside and the pooled gap falls to +0.5 pp [−0.4, +1.4]; over the five admitted routes it is −0.5 pp [−1.3, +0.4]; and Claude Opus 5 and Claude Sonnet 5 are at exactly 0.000 in all thirty of their races each. These intervals resample races without stratifying by route, so the analyser's pooled interval is a little wider than the figure's, which resamples inside each route; the point estimates agree exactly. The scripted-rival campaign agrees independently. There the rival is fixed and the route's seat is counterbalanced five and five inside every cell, and the gap over its 36 cells, pooled across the three routes it now covers, is −1.5 pp [−3.7, +0.7].

That turns an unexplained asymmetry into a result, and it points the same way as the rest of the paper rather than against it. The mechanism is symmetric exactly where the gap is largest: at round one every state field is zero, and the regenerating script checks that all 270 opening prompt pairs are byte-identical under a `Company_1`/`Company_2` name swap rather than assuming it. So what is left to respond to at that point is an ordinal name token, and the route that responds to the name instead of to the state is a route the comprehension gate refused. Measured the way its risk response is measured, decision-weighted, GPT-5.4 nano's whole-race seat gap of 19.4 pp is exactly twice its risk response of 9.7 pp, and at the opening, the one place where the two prompts are identical and the contrast is therefore a pure label effect, its gap is 30.0 pp [+13.3, +46.7]. States diverge after round one, so only the opening figure is a clean label contrast; the whole-race figure is the one that is comparable with the risk response. The opening is also where the pattern stops belonging to the refused routes alone, and that belongs here rather than in a footnote: two admitted routes carry an opening gap whose interval excludes zero, +33.3 pp [+16.7, +50.0] on Claude Sonnet 5, which never once opened UNSAFE from the first seat, and +16.7 pp [+3.3, +30.0] on GPT-5.4. Four of the nine routes open UNSAFE from both seats in every race and can show no gap at all. So the screen separates the routes on the whole-race asymmetry and does not separate them at the opening, and the design protection the paper actually relies on is the seat counterbalancing rather than the screen. The usual bounds hold. This is nine endpoints in one game at one prompt version, nothing is randomised except the chair, and a route whose measured gap is zero is at zero because its two seats agree, not because the design removed the effect.

## A.24 Where a race goes, once it gets somewhere

Every behavioural number reported elsewhere in this paper describes one seat: it says what a route plays, given the risk it was told about or given what it saw its rival do. A race is not one seat. What happens to the world in a round is the *joint* outcome, and with two seats and two actions that outcome is one of three states: both safe, one each, and both racing. A race is then a walk on three states, which is the object the evolutionary lane of this paper reasons about and the object the behavioural lane did not have. Figure 13 builds that walk from the same nine-route neutral corpus used throughout: 270 races, 2,511 rounds, and 2,241 transitions between consecutive rounds of one race.

The two mixed outcomes are collapsed into a single *split* state. These are self-play races, so the two seats are the same route under the same prompt and differ only by which chair the protocol sat them in, which makes `Company_1` unsafe and `Company_2` unsafe the same event under a relabelling. The regenerating script tests that exchangeability rather than asserting it: `Company_2` is the racing seat in 53.5% of the 818 split rounds, close enough to a coin that a four-state chain would buy noise rather than structure. Transitions are counted only between consecutive rounds of the same race, and the last round of a race is dropped because it has no successor. Each race draws its own horizon once, before the first move, so censoring cannot depend on where the walk had got to.

Three readings survive the figure. First, restraint is not sticky. Pooled over the eight routes that ever changed state inside a race, mutual restraint is followed by mutual restraint on only 34.4% of occasions, 105 of 305, and in seven of the nine routes the chance that restraint holds into the next round is below one half. Second, only `anthropic/claude-opus-5@default` spends most of its rounds in mutual restraint, 66.7%, and it is also the one route whose two seats never once played differently. Third, the road back matters: a split is repaired on 5.8% of the rounds that follow one, 42 of 723, while both seats stand down together on 18.3% of the rounds that follow mutual escalation, 176 of 964. Of the 218 arrivals at mutual restraint, 80.7% are both seats standing down at the same moment rather than one seat stepping down and the other joining it a round later.

The Opus 5 matrix deserves its own sentence, because a pooled matrix with every count on the diagonal invites the word *frozen*, and that word would invert this paper's own reading of the route. The constancy is within a race and not across the design. At $p_r^{\max} = 0.1$ all ten of its races sat in mutual escalation for every round; at 0.6 and at 0.9 all twenty sat in mutual restraint. The route therefore moves further with risk than any of the other eight, a full 100 points of UNSAFE rate against 57 for the next most responsive. It is a step in self-play: deterministic inside a cell, and the one route in the figure whose pooled matrix is an artifact of the pooling rather than a summary of it.

*What it does not show.* A transition count is not a mechanism. Nothing here says a route noticed the state, and the same matrix would be produced by two seats reacting only to their own accumulated private risk. Rows pool the three risk levels, and risk moves these dynamics hard, so a route's matrix summarises three regimes rather than stating a law. The ordering in panel c is a property of the pooled rows and not of every regime inside them: at $p_r^{\max} = 0.1$, where mutual restraint is reached at all in only 4% of rounds, the two roads swap and repair from a split becomes the likelier one, 7.1% against 3.8%. Thirty races per route is enough to separate a route that never moves inside a race from a mobile one and not enough to pin a single cell to a point. And a state is only a joint action, so two routes can walk the same three states with entirely different progress, payoff and accumulated private risk behind them.

## A.25 The designed boundary at round five

The mechanism has one sharp, designed, disclosed edge in it, and it sits between round four and round five. The prompt every player reads says that the race lasts at least five rounds and that from round five onward each completed round ends the race with probability 20%. The continuation probability is therefore exactly 1 while the race is short of round five and 0.8 from round five onward. The rule is disclosed; the realised length is not. This is the one quantity in this game that the folk-theorem tradition says should move cooperative play, because a threat certain to be carried out sustains cooperation that a threat with a one-in-five chance of never arriving does not. Figure 14 asks whether the routes' play moves there. The regenerating script parses the horizon sentence out of every prompt in the corpus rather than trusting the protocol document, so it fails rather than drawing a boundary the players were never told about.

One route steps up at the boundary and one steps down. Comparing round five against rounds one to four, the window in which all thirty races per route contribute by construction, `openai/gpt-5.4-2026-03-05` plays 24.6 points more UNSAFE, with a 95% percentile interval of 16.2 to 32.5, and `google/gemini-3.5-flash-lite` plays 17.1 points less, with an interval of −28.8 to −4.6. The other seven intervals cover zero, several of them wide enough to hold a twenty-point move in either direction. `anthropic/claude-opus-5@default` plays one rate in every round of the corpus, so its step is exactly zero and could not have been anything else.

*Why this is a comparison and not a discontinuity estimate.* Two confounds sit between the design and any causal reading, and the figure carries both on its face rather than in a footnote. The first is accumulation: a later round is also a round with more history behind it and with more accumulated private risk, and because those move together inside a race no contrast between "before round five" and "round five onward" can separate them from the horizon. That is why panel b is labelled as a step across round positions, and why panel c exists at all. Comparing each race with itself one round later holds accumulation nearly fixed: every rung of that ladder is one round of extra history, and exactly one rung crosses the boundary. The second is survivorship. Races have different sampled lengths, so the races that reach round twelve are not the races that reached round three. Every route was run against the same ten horizon draws at each of the three risk levels, so no route is favoured, but the race set still changes along the axis. The filled estimate in panel b and the first four rungs of panel c stay inside rounds one to five, which every race reaches by construction; the last two rungs rest on the twenty-four races per route that get that far, and the panel says so.

The ladder is where the argument lands. The rung that crosses the boundary is the only rung after the opening two whose interval clears zero, so it is not nothing: −6.5 points, interval −11.1 to −1.9. But it is small beside the opening move, −39.3 points from round one to round two, and beside the +25.4 points from round two to round three, and it is negative, which is the direction opposite to the one a shortening shadow of the future is supposed to push. The grey context in that panel is not decoration: three of the nine route

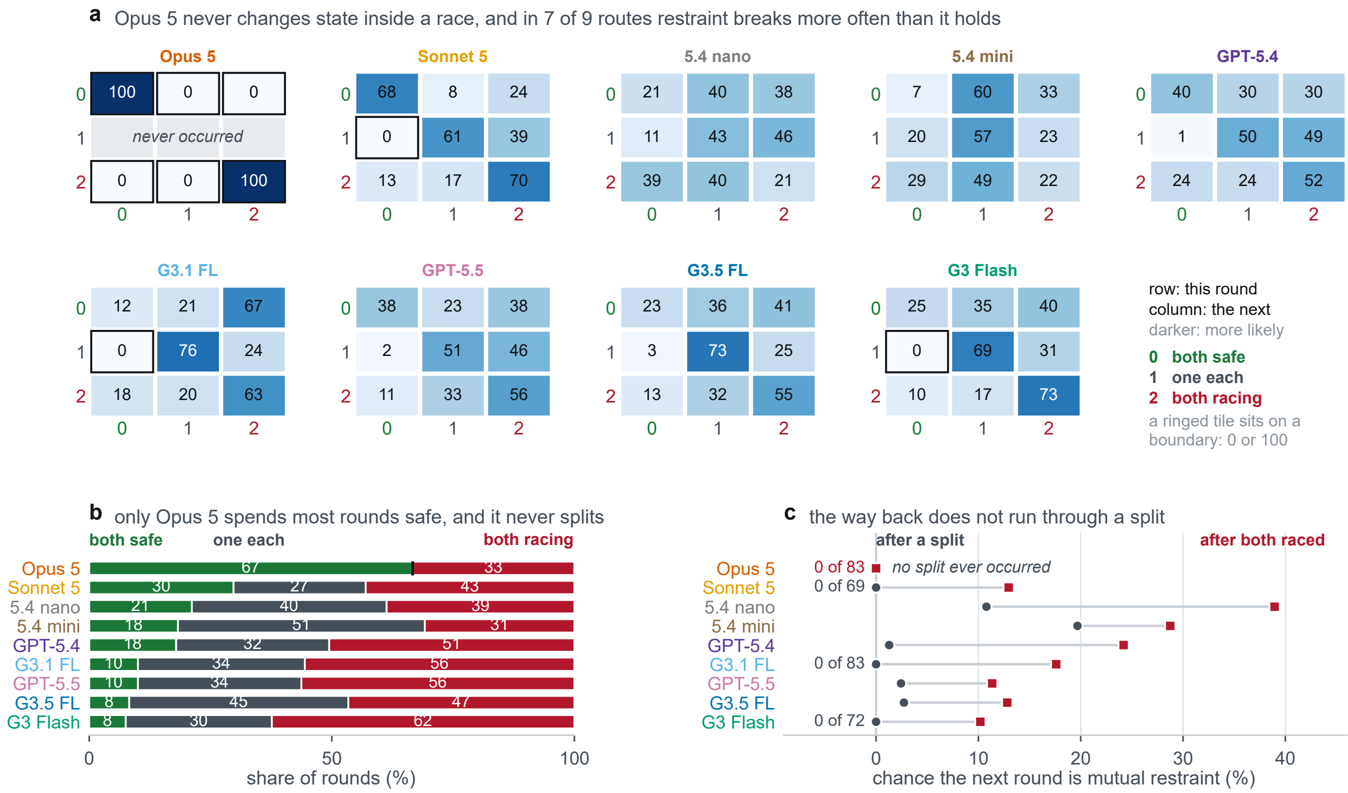



**Figure 13: The race as a walk on joint outcome states. The joint outcome of a round is how many of the two seats played UNSAFE, giving three states: both safe, one each, and both racing. Panel a gives one transition matrix per route, rows this round and columns the next, each row summing to 100, ordered from the route that spends most of its rounds in mutual restraint to the route that spends least. A ringed tile sits exactly on a boundary of 0 or 100 rather than near one, and the grey row for Claude Opus 5 marks a state that never occurred rather than a value that is missing. Panel b divides each route's rounds between the three states. Panel c draws the two roads back to mutual restraint against each other: pooled over the eight routes that ever changed state inside a race, a split is repaired on 5.8% of the rounds that follow one and both seats stand down together on 18.3% of the rounds that follow mutual escalation, so four arrivals at restraint in five are simultaneous. The figure is not a mechanism: a transition count cannot say that a route noticed the state, and the same matrix would follow from two seats reacting only to their own accumulated private risk. Rows pool the three risk levels, which move these dynamics hard. Both seats of every race are the same route, so every number here is what a route does against a copy of itself.**

means point the other way, so the pooled interval is a statement these nine routes make together rather than one that each of them makes, and resampling routes rather than races widens it to cover zero.

*What it does not show.* This is not a regression discontinuity. The running variable is a round index that also carries history, the design has no observations arbitrarily close to the threshold, and nothing here is a causal estimate of the horizon. Nor is it a test of whether these routes can reason about a horizon at all; it tests whether their play moves when the stated horizon changes, in this game, at this prompt version. The absence of a step is evidence about the step and not proof that the prompt sentence went unread: panel a shows one route whose play is a constant, and a constant cannot move at a boundary. Both seats of every race are the same route, so what is measured is a horizon response against a copy of itself and not against a different opponent. Nine commercial endpoints are not a sample from a population of models.

## A.26 Three audited routes against a rival they cannot influence

Every gameplay result reported elsewhere in this paper is self-play: both companies in a race are the same endpoint. That design cannot separate a route responding to its rival from a route running through a phase of its own, because the rival is the route. It also leaves this paper's two lanes disconnected. The evolutionary lane is built on four reduced strategies, *Always Safe*, *Always Unsafe*, *Conditional Safe* and *Conditional Unsafe*, and until this campaign the empirical lane had never played against any of them.

*What was collected.* Three gate-admitted routes, Gemini 3 Flash, GPT-5.4 and Claude Sonnet 5, whose full route identifiers are in

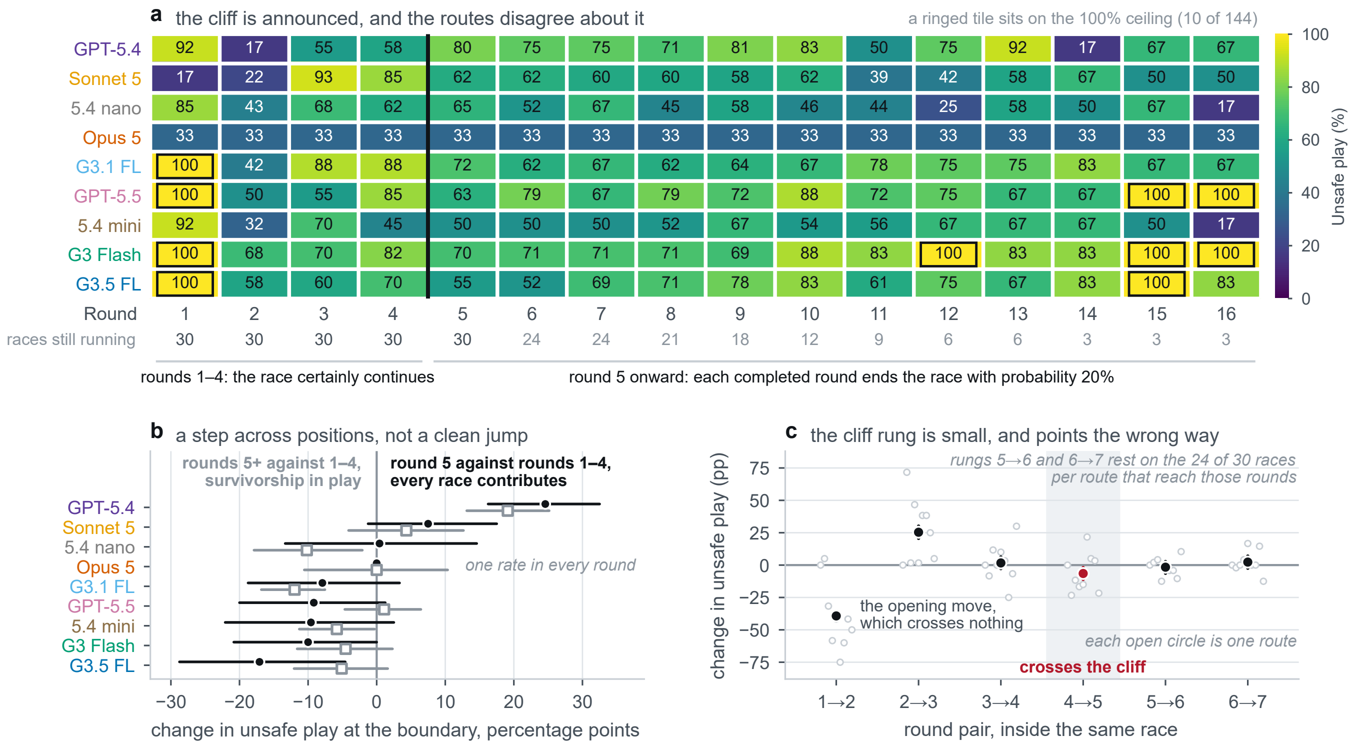



**Figure 14: UNSAFE play by round position, with the designed boundary at round five. Panel a gives one row per route and one column per round, with the surviving race count printed under each column and a ring on the tiles that sit on the 100% ceiling, where the contrast above is truncated rather than measured. Panel b gives the step at the boundary per route under two windows: round five against rounds one to four, where all thirty races per route contribute, and all later rounds against rounds one to four, where survivorship and unequal decision counts are in play. Seven of the nine intervals cover zero and the two that do not point in opposite directions, +24.6 points for GPT-5.4 and −17.1 for Gemini 3.5 Flash Lite. Panel c is the placebo ladder: within-race change from one round to the next, pooled over routes, with the nine route means as grey context. *This is a comparison across round positions and not a discontinuity estimate.* A later round also carries more accumulated history and, because private risk is the fraction of a player's own completed actions that were UNSAFE, more accumulated private risk; round position and accumulated state are collinear inside a race, so no contrast between the two sides of the boundary can separate the shortening horizon from what a later round otherwise brings with it. The rung that crosses the boundary is −6.5 points beside an opening move worth 39.3, and its sign is the opposite of the one a shortening shadow of the future is supposed to push.**

Table 1, each played the same game against each of those four strategies at each of the three private-risk levels. That is 36 cells, ten repetitions each, 360 races and 3,348 route decisions, 93 route decisions per cell, with zero parse failures and zero deviations by any scripted rival. The rival is executed by the task file rather than called, so it costs no requests and cannot be wrong about its own strategy for any reason a model could be.

Playing a language model against fixed strategies is established practice rather than something introduced here: Akata et al. [2] run always-cooperate, always-defect and defect-once agents against five models, and Payne and Alloui-Cros [10] run Tit-for-Tat and Grim Trigger against frontier models under a varying termination probability. Our two conditional rivals are theirs. What is particular to this campaign is that the rival is never disclosed, that its every move is replayed and checked after the fact, that the route's seat is counterbalanced, and that the same frozen design is run on more than one endpoint.

Three properties of the design are load-bearing and were asserted offline before any of it was pushed. The prompt is byte-identical to the neutral baseline's and never says the rival is scripted, so a race here is comparable with a race there; the rival is a strategy, not a disclosure. The route's seat is counterbalanced, five repetitions in each seat per cell, because this study has measured a seat effect in the neutral baseline, on prompts that differ only in which of two names is whose, and a fixed seat would fold that effect into every number. And a conditional strategy answers the *route*'s previous move rather than its own, which is the property that separates a rival from a mirror: a strategy answering its own history would be a constant, and the whole conditional arm would measure nothing.

*How the rival was audited.* A scripted rival is code, so it cannot be audited by reading a response. It was audited by replay: `scripts/ingest_scripted_cell.py` recomputes every rival move from the route's own recorded moves and refuses the cell on a single deviation. All 36 cells replayed clean. This is the one failure that would have left the data looking perfectly healthy while answering a different question than the one asked.

*The collection record.* The extension to two further routes was planned and committed before the first of its cells was started, and the plan fixed which account would collect which cell. No account carries enough request budget for a whole route, so the workload is partitioned into the analysis unit itself, one route by rival by risk cell, and each cell is collected whole on one declared account. Table 24 is the record of what actually ran, and it matches the declared assignment cell for cell. Within a route, each of the four strategies was collected on three different accounts and no account collected the same strategy twice, so an account effect would show up as an inconsistency between cells rather than hide inside a contrast. Across the two routes added here, no cell was collected on the same account, so the comparison between them is not confounded with the account. Against the route the campaign began on, whose four-account rotation was fixed earlier and could not be chosen to avoid this, two of the 24 later cells repeat its account: *Always Unsafe* at $p_r^{\max} = 0.9$ on Claude Sonnet 5 and *Conditional Safe* at $p_r^{\max} = 0.1$ on GPT-5.4. All 36 cells carry the protocol identifier `ai-race-scripted-opponent-v1` and the prompt version `ai-race-fairgame-v3`, and the three recorded task hashes are one per risk level and identical across the three routes, which is what shows the mechanism did not move between them.

*An attempted completion that quota stopped.* The design was declared for the last two admitted routes and then stopped by account budgets after a single cell, and both the plan and that cell are in the artifact, so the outcome belongs here rather than being left to be inferred. Twenty-four further cells were fixed in advance, twelve on GPT-5.5 and twelve on Claude Opus 5, each named with the account that would collect it. One landed: Claude Opus 5 against *Always Safe* at $p_r^{\max} = 0.6$, ten races, 93 route decisions, zero parse failures, the rival replayed with zero deviations, and the contract hash that risk level requires. Four of the five accounts then returned quota refusals and were retired by the plan's own stopping rule; a fifth failed three times for a different reason, which the record keeps separate because it was not a refusal of budget; and the remaining fifteen cells were never started, because the account that owned each of them had already refused. No cell was moved to another account, since choosing a new account after a refusal would select on the outcome. The failure record is retained in the artifact with all nine attempts, and no race completed in any of them.

*What the one cell does not decide.* It reads 1.1% UNSAFE, one decision in 93, and Claude Opus 5's self-play rate at the same risk level is 0.0% across 186 decisions. Both sit on the floor, so the cell cannot separate them, and the two arms that would have separated a policy from a mirror match, *Always Unsafe* and *Conditional Unsafe*, were never collected. Whether Claude Opus 5's self-play step is its policy or an artifact of meeting a copy of itself therefore remains open, and nothing in this paper claims otherwise. The scripted-opponent campaign covers three of the five admitted endpoints, exactly as it did before this attempt.

*Finding one: the shape replicates.* Every route plays UNSAFE least against the rival that always plays SAFE and most against the rival that never does, with the two conditional rivals in between, and it does so at every risk level. The ordering is strict in eight of the nine route by risk cells. The ninth is not a counterexample but a saturated measurement: at $p_r^{\max} = 0.1$ Gemini 3 Flash plays UNSAFE on all 93 of its decisions against *Always Unsafe* and on all 93 against *Conditional Unsafe*, and two cells with no room above them cannot be put in an order. The paired rival-stance contrast, the rate against *Always Unsafe* minus the rate against *Always Safe*, is positive in all nine cells, and the smallest lower bound anywhere is +35.8 points, on GPT-5.4 at $p_r^{\max} = 0.9$. A scripted rival cannot be influenced by the route, so the direction of that contrast is not in question.

The response to stated risk survives as well, once the rival is held fixed. Against *Always Safe* the rate never rises with risk on any route: it falls from 24.7% to 14.0% on Gemini 3 Flash and from 36.6% to 25.8% on GPT-5.4, while on Claude Sonnet 5 it falls from 22.6% to 10.8% and then stays there, so the response is strictly falling on two of the three routes and flat between the two higher risk levels on the third.

*Finding two: the magnitude does not replicate, and that is the more informative result.* How far the rival's stance moves a route is a property of the checkpoint. Gemini 3 Flash answers its rival by about seventy points at every risk level, +73.5, +67.4 and +71.7. GPT-5.4 answers by about fifty, +47.5, +48.0 and +49.8, and its interval does not meet Gemini 3 Flash's at any of the three risk levels. Claude Sonnet 5 gives a Gemini-sized answer at the two lower risk levels, +63.8 and +65.7, and then falls to +45.0 $(42.0, 48.2)$ at $p_r^{\max} = 0.9$, where its interval no longer meets Gemini 3 Flash's +71.7 $(65.6, 77.5)$. These are three separate samples, so intervals that fail to overlap are a conservative sign of a difference rather than a test of one.

The level differs as well as the swing. GPT-5.4 carries the highest rate against *Always Safe* at every risk level, 36.6%, 32.3% and 25.8%, so it is the one route here that still plays UNSAFE about a third of the time against a rival that never does; the other two fall to between 10.8% and 24.7%. Responding to the rival is therefore the general finding across these three endpoints, and how strongly a route responds is specific to the checkpoint. That is the same conclusion this paper reaches from the admission gate and from the neutral baseline, reached here through a design in which the rival cannot answer back.

*We name the contrast for what it compares.* Calling the safe-rival arm exploitation would assert that the route takes the opportunity a non-punishing rival offers, and the measured rates say the opposite: on every route that arm carries the lowest rates anywhere in the campaign, below what the same route plays against itself. What the routes do is match the rival, keeping most of their restraint against a rival that keeps its own and abandoning much of it against one that never had any. A strategy name is not a summary of what the rival played either. *Conditional Safe* opens SAFE and then mirrors,

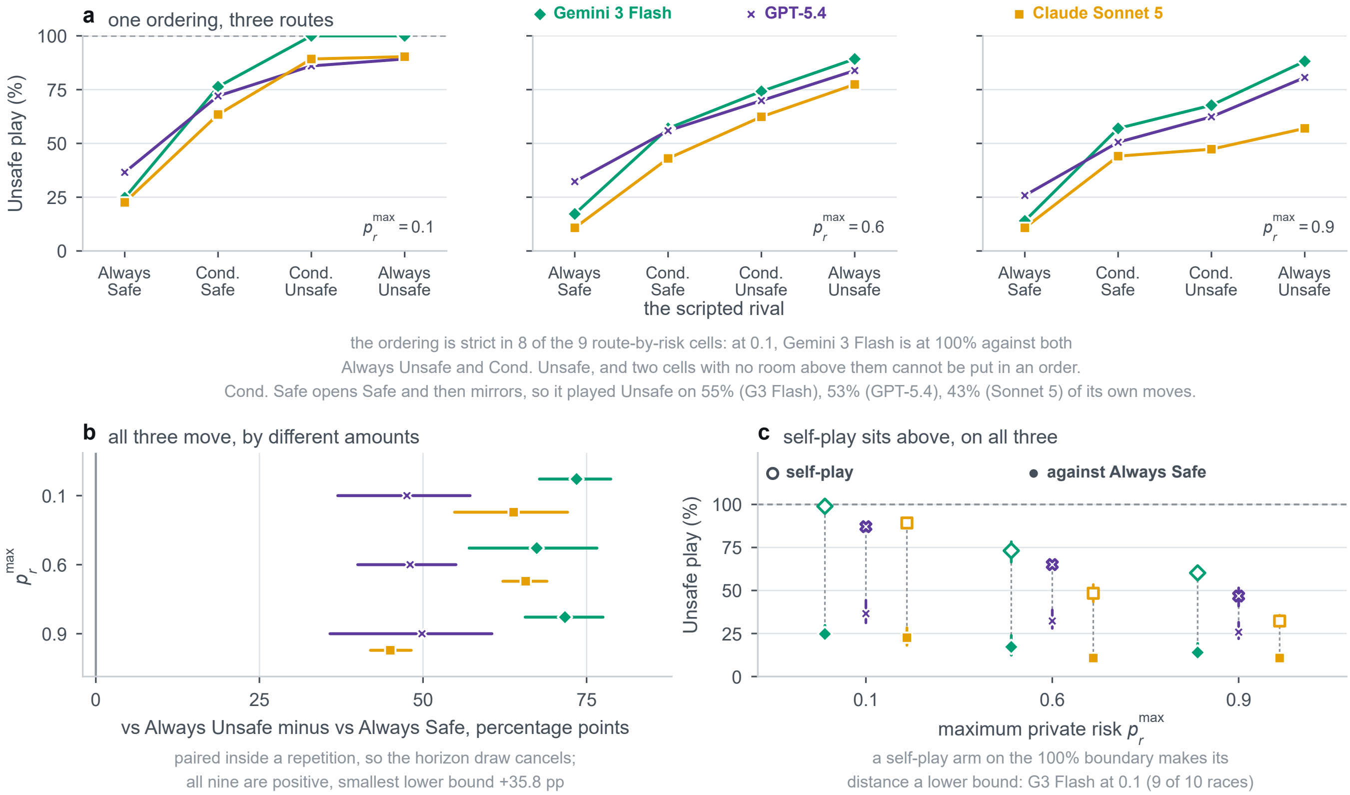



**Figure 15: The scripted-opponent campaign on three gate-admitted routes. (a) The ordering, one facet per stated risk: four rivals across, the route's own UNSAFE play up. A rising line is the ordering, and the vertical distance between the three lines is how far the routes differ in level. (b) The rival-stance contrast, the rate against *Always Unsafe* minus the rate against *Always Safe*, differenced inside a repetition so that the horizon stopping draw is removed. All nine intervals lie to the right of zero, and the three routes separate. (c) The comparison the tables cannot make: each route's rate against a rival that stays SAFE, as a filled marker, beside the rate the same route reaches when its rival is a second copy of itself, as an open marker.**

so it played UNSAFE on 55.2%, 53.4% and 43.0% of its own moves against the three routes, in the order of the columns above.

*The rival's opening move.* *Conditional Safe* and *Conditional Unsafe* are the same strategy from round two onward, both repeating the route's previous move, so their difference isolates the rival's first move and nothing else. It is worth +25.1, +15.8 and +11.5 points on Gemini 3 Flash, +14.1, +13.9 and +11.0 on GPT-5.4, and +25.0, +19.6 and +3.4 on Claude Sonnet 5. Eight of those nine intervals exclude zero. The ninth, Claude Sonnet 5 at $p_r^{max} = 0.9$, has a lower bound of exactly 0.0 over its ten blocks and therefore does not, which is reported here rather than rounded away. On Gemini 3 Flash at $p_r^{max} = 0.1$ a rival that merely opens UNSAFE and then mirrors drives the route to every one of its 93 decisions being UNSAFE, the same as a rival that is UNSAFE throughout.

The route's own opening move is not a constant across the three routes, and no reading here depends on it: Gemini 3 Flash opened UNSAFE in all 120 of its races, GPT-5.4 in 116 of 120, and Claude Sonnet 5 in only 23 of 120, all of them at the lowest risk level. The opening-move contrast is about the rival's first move, which the task file fixes, and not about the route's.

*What it means for the rest of this paper.* A self-play rate is not a measurement of how a route treats risk. On all three routes and at all three risk levels, the rate against a fixed and safe rival sits below the rate the same route reaches in the neutral baseline, where its rival is a second copy of itself. The distance is 74.2, 55.9 and 46.2 points on Gemini 3 Flash, 50.5, 32.8 and 21.0 on GPT-5.4, and 66.7, 37.6 and 21.5 on Claude Sonnet 5. The first of those is a lower bound, because nine of the ten Gemini 3 Flash self-play races at $p_r^{max} = 0.1$ have every decision UNSAFE and that arm has no room above it. A self-play rate is an equilibrium of one policy meeting itself, and it says more about what happens when a policy meets a copy of itself than about what the policy does with risk. Every self-play number in this paper should be read that way.

That self-play can hide behaviour is not a new observation. Lekeas and Stamatopoulos [7] report phenomena that are, in their words, invisible in self-play, including that which agent moves first decides which equilibrium a pair reaches. The contribution here is narrower and specific to this setting: what self-play hides is the *risk response* itself, the one quantity this game exists to elicit, and it hides it on every route we have run this design on.

**Table 23: Each route against each scripted rival. Ten races and 93 route decisions per cell, both seats used in every cell, zero parse failures and zero rival deviations anywhere. Rates are per cent of the route's own decisions. Contrasts are differenced inside a repetition, so the horizon stopping draw is removed from the comparison, with the 95% percentile interval taken over the ten blocks; the seed pairing was verified for all ten in every one of the eighteen contrasts. Ten blocks make the resampling distribution coarse, so an interval endpoint carries a few tenths of a point of Monte Carlo movement and is quoted to one decimal for consistency rather than because it is resolved that finely. The two cells at 100.0% sit on a boundary, and every contrast through them is truncated rather than complete. The self-play rate is the same route in the neutral baseline of §A.3, 10 races and 186 decisions per cell, and it is a different design rather than another rival: at risk 0.1 nine of Gemini 3 Flash's ten self-play races have every decision UNSAFE, so the 74.2 below it is a lower bound on that distance.**

| | Gemini 3 Flash | | | GPT-5.4 | | | Claude Sonnet 5 | | |
|---|---|---|---|---|---|---|---|---|---|
| $p_r^{\max}$ | 0.1 | 0.6 | 0.9 | 0.1 | 0.6 | 0.9 | 0.1 | 0.6 | 0.9 |
| Always Safe | 24.7 | 17.2 | 14.0 | 36.6 | 32.3 | 25.8 | 22.6 | 10.8 | 10.8 |
| Conditional Safe | 76.3 | 57.0 | 57.0 | 72.0 | 55.9 | 50.5 | 63.4 | 43.0 | 44.1 |
| Conditional Unsafe | 100.0 | 74.2 | 67.7 | 86.0 | 69.9 | 62.4 | 89.2 | 62.4 | 47.3 |
| Always Unsafe | 100.0 | 89.2 | 88.2 | 89.2 | 83.9 | 80.6 | 90.3 | 77.4 | 57.0 |
| Rival's stance | +73.5 | +67.4 | +71.7 | +47.5 | +48.0 | +49.8 | +63.8 | +65.7 | +45.0 |
| 95% CI | (67.8, 78.7) | (57.1, 76.6) | (65.6, 77.5) | (37.0, 57.2) | (40.0, 55.0) | (35.8, 60.5) | (54.8, 72.1) | (62.2, 68.9) | (42.0, 48.2) |
| Opening move alone | +25.1 | +15.8 | +11.5 | +14.1 | +13.9 | +11.0 | +25.0 | +19.6 | +3.4 |
| 95% CI | (20.9, 29.7) | (8.3, 23.5) | (4.5, 19.7) | (6.0, 22.7) | (8.9, 18.1) | (6.6, 15.5) | (20.9, 29.3) | (17.7, 21.5) | (0.0, 7.1) |
| Self-play rate | 98.9 | 73.1 | 60.2 | 87.1 | 65.1 | 46.8 | 89.2 | 48.4 | 32.3 |
| minus Always Safe | 74.2 | 55.9 | 46.2 | 50.5 | 32.8 | 21.0 | 66.7 | 37.6 | 21.5 |

**Table 24: The collection record: which account collected which cell. The assignment was fixed in a collection plan committed before the first cell of the extension was started, and this table is read back from each cell's own receipt rather than from the plan. The five accounts are anonymised for double-blind review and appear here as A to E, labelled in order of first appearance in this table; the mapping is retained privately and is restored in the camera-ready artifact. An account is a billing boundary and not a model boundary: route, prompt, parser, protocol, temperature request and seed structure are identical in every cell. The rotation is what makes that assumption checkable instead of merely asserted.**

| | Gemini 3 Flash | | | GPT-5.4 | | | Claude Sonnet 5 | | |
|---|---|---|---|---|---|---|---|---|---|
| $p_r^{\max}$ | 0.1 | 0.6 | 0.9 | 0.1 | 0.6 | 0.9 | 0.1 | 0.6 | 0.9 |
| Always Safe | A | B | C | B | C | D | E | A | B |
| Conditional Safe | C | E | A | C | D | E | A | B | C |
| Conditional Unsafe | E | A | B | A | B | C | D | E | A |
| Always Unsafe | B | C | E | E | A | B | C | D | E |

*What it does not show.* Three routes, one game, one prompt version, ten races per cell. Three commercial endpoints are not a sample from a population of models, and four strategies are four points in a space of policies rather than a sample of opponents, so a scripted rival establishes what a route does against *that* rival. Three routes are enough to say the ordering replicates and that the size of the response does not; they are not enough to characterise endpoints in general, and a fourth route could differ from all three. The design is causal in the narrow and precise sense that the rival cannot be influenced by the route, so the direction of the effect is not in question; it is not causal about anything outside this game. Nothing here licenses a claim about strategy, intent, or reasoning: these are conditional frequencies. It is regenerated by `scripts/analyze_scripted_opponent.py` with `--all-routes`, which takes one endpoint at a time, writes one derived table per endpoint rather than pooling them, and refuses a cell that did not complete, that carries a parse failure, that is the wrong protocol, that has the wrong number of races, whose seat counterbalance is not five and five, or whose rival deviated from the strategy its directory claims.

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